

$$y = 2x + 3$$

$$\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

$$2(x - 3) = 14$$

★ VOLUME 2 ★

EMERGENCY SUB PLANS

for
MIDDLE SCHOOL MATH

GRADES
6-8



**NO-PREP
ACTIVITIES**



**READY-TO-USE
RESOURCES**



**PERFECT FOR
EMERGENCY
LESSONS**



**GREAT FOR
SUBSTITUTE
TEACHERS**



**PRINT OR
DIGITAL USE**

KEEP LEARNING ON TRACK WHEN PLANS CHANGE

RATIO & PROPORTIONS

Solve each problem.

- The ratio of boys to girls in a class is 3:5. If there are 24 girls, how many boys are in the class?



- Simplify the ratio 18:24.

SOLVE FOR x

Solve each equation.

- $2x + 5 = 17$

- $3(x - 4) = 21$

- $\frac{x}{4} + 3 = 8$

- $5x - 7 = 28$

PLOT & ANALYZE

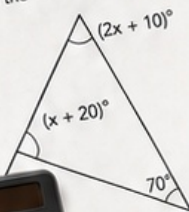
The graph shows the number of skateboards sold each week.



- How many skateboards were sold in Week 4?
- Between which weeks did sales decrease?

GEOMETRY

Find the value of x.



x =

**MATH
EMERGENCY
KIT**

ORDER OF OPERATIONS

Evaluate each expression.

- $12 + 6 \times 2 - 4$
- $(16 - 8) + 3 + 7$
- $2^3 + 3 \times (8 - 5)$
- $24 \div (2 + 2) + 5$

YOUR CLASSROOM LIFESAVER



**MADE BY
TEACHERS**
for

Set this up while you are well

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a morning when you are not.

The one job you have today

Print pages 3–6 and leave them in a clearly labeled folder where a substitute will find them — top drawer, on the desk, or at the front office. Everything a sub needs is in those four pages.

You do not need to print the lessons ahead of time. The sub picks one on the day and copies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a folder or binder. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. Those are the pages a substitute reads first.
- 3 **Tell your department chair where it lives.** On a real sick day you will not be answering the phone, so somebody else needs to know.
- 4 **Cross off lessons as you use them.** There is a tracker on page 4. Two years of absences will never repeat if you keep it current.
- 5 **Optional: leave a class roster** and a note about students who need a reader, extra time, or specific seating.

Why nothing here needs technology

On a sub day the Chromebook cart is often locked, the login server is down, or nobody knows the room code. Every lesson runs on paper and a pencil.

Calculators are optional throughout, and each lesson says where one actually helps.

Why it is not busy work

Every lesson targets a real Grade 6–8 standard and produces graded evidence you can put in a folder.

Away for a week? Running four of these in sequence is a defensible unit, not a holding pattern.

About the answer keys

Every key sits on the **last page of its lesson** and shows the **working**, not just the answer. That is what lets a sub who has not taught math in years respond when a student asks “*but why?*”

If you also own the other volumes

Volume 1, Volume 3, Volume 4 contain thirty further lessons and not one of them repeats a topic from this pack. Keep the folders side by side and a substitute can cover weeks without a student seeing the same sheet twice.

You do not need to be a math teacher

You have walked into a middle school math class. You may not have done algebra since high school. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to be able to do the math. Every lesson is a paper worksheet with a complete answer key that shows the working. Your job is to hand it out, keep the room settled, and check answers against the key. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for Grades 6, 7 or 8. If you cannot decide, use Lesson 08.
- 2 **Copy the student pages only.** Each lesson tells you exactly which page numbers. One per student.
- 3 **Keep the teacher page and the answer key.** Do not copy those.
- 4 **Read the teacher page once** before the bell. Two minutes, and it includes the exact words you can use to start.

If a student says “we already did this”

Ask them to prove it by finishing the challenge section on the last page. If they genuinely have, switch them to a different lesson — there are ten, and they will not have done all of them.

If they finish early

Every lesson has a challenge task on its final student page. There is also a discussion question on the teacher page you can run with the whole class for the last five minutes.

What to leave behind

Collect the sheets. Fill in the handover note on the last page of this pack — ninety seconds, and it tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	The Number Detective	Factors, GCF & LCM	★★	55	7–9
02	Slices and Shares	Fraction operations	★★★	55	12–14
03	Scale It Up	Proportion & scale drawings	★★	55	17–19
04	Powers of Ten	Exponents & scientific notation	★	50	22–24
05	Same Thing, Shorter	Expressions & like terms	★★	55	27–29
06	More Than, Less Than	Inequalities	★★★	55	32–34
07	Boxes and Cans	Volume & surface area	★★	60	37–39
08	Angles Everywhere	Angle relationships	★	55	42–44
09	The Diagonal Problem	The Pythagorean theorem	★★	55	47–49
10	Rate of Change	Slope & linear relationships	★★★	60	52–54

Easiest to supervise

04, 07 and 08. Concrete and visual. A wrong answer usually looks wrong, so students catch themselves.

Best for a quiet class

02, 05 and 09. Steady practice with definite answers. Fast finishers will race — that is what the challenges are for.

Best for a restless class

01, 03 and 06. Puzzles, scale drawings and real budgets. Noisier, but productively so.

Used-lesson tracker • for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Factors, GCF & LCM			
02 — Fraction operations			
03 — Proportion & scale			
04 — Exponents			
05 — Expressions			
06 — Inequalities			
07 — Volume & surface area			
08 — Angles			
09 — Pythagoras			
10 — Slope & linear graphs			

The first ten minutes

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in and take a seat. Bags off the desks. You will not need a Chromebook today — everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is out today. My name is _____. This is a real math lesson, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their own teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the first panel together	Have a student read the worked example aloud
10–40	Students work through the tasks	Circulate. Check against the answer key
40–50	Challenge task for early finishers	Point them to the last student page
50–55	Collect sheets, pack up	Fill in the handover note

"I don't get it"

Almost always means *I don't know how to start*. Point at the worked example on the first page and ask them to copy its first line with their own numbers.

That single move unsticks most students without you needing to explain anything.

"Can I use a calculator?"

Unless the teacher page says otherwise, yes for the arithmetic and no for the reasoning tasks. Each teacher page states which.

If in doubt, allow it. A student doing the thinking with a calculator is better than one doing nothing.

Copying reminder

Student pages only. Each teacher page gives the exact range. The answer key is always the last page of the lesson — keep it.

The Number Detective

READ BEFORE THE BELL

2 MINUTES

Students break numbers into their prime factors, then use those factors on GCF and LCM questions that arrive disguised as party bags, bus timetables and ribbon. The factoring is the easy part — deciding which of the two a question is asking for is the real skill.

You do not need to know this subject

Every answer is a whole number and the key shows the working for each one. If you can check that two numbers match, you can grade this lesson.

Photocopy these pages only

Pages 7–9

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Calculators are not needed and are best left in the bag.

WHAT TO SAY TO START

“Every whole number bigger than 1 is built out of primes, and there is exactly one way to build each one. Today you find the fingerprint of a number — and then use it to answer questions that look nothing like factoring.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the fingerprint panel	Do the factor tree for 60 on the board
12–24	Task 1.1 — prime factorisation	Circulate. Every answer is in the key
24–36	Task 1.2 — GCF and LCM	The pattern question at the end is the payoff
36–48	Task 1.3 — the word problems	Which one does the question want?
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is 1 a prime number?”

No. A prime has exactly two factors, 1 and itself. The number 1 has only one.

“Which is bigger, the GCF or the LCM?”

The LCM. The GCF can never exceed the smaller number; the LCM can never be below the larger one.

“How do I know which one the question wants?”

Cutting something up or sharing it out means GCF. Two things repeating until they line up means LCM. Task 1.2b drills exactly this.

If they finish early

Task 1.5 asks for the second number when only the GCF, the LCM and one number are known. Beyond that, ask them to find two numbers whose GCF is 1 and say what that means about the pair.

Curriculum focus • for the regular teacher’s records

Grade 6: greatest common factor, least common multiple and prime factorisation of whole numbers. Discussion question for the last five minutes: “Every number has exactly one prime fingerprint. Why is that useful?”

One number, one fingerprint

UNDERSTAND

14 MIN

NAME _____

CLASS _____

DATE _____

Every number has a fingerprint

A **prime** has exactly two factors: 1 and itself. Everything else breaks down further, and eventually every number lands on primes.

The fingerprint is unique. Wherever you start splitting, you always finish with the same set of primes.

2 3 5 7 11 13 17 19 23 29

$$\begin{aligned}
 60 &= 6 \times 10 \\
 &= (2 \times 3) \times (2 \times 5) \\
 &= 2^2 \times 3 \times 5
 \end{aligned}$$

Start differently:

$$\begin{aligned}
 60 &= 4 \times 15 \\
 &= (2 \times 2) \times (3 \times 5) \\
 &= 2^2 \times 3 \times 5 \quad \text{same}
 \end{aligned}$$

There is no wrong place to start. That is the point of the lesson — the ending is fixed.

Task 1.1 • Break it down

DO

Write each number as a product of primes. Use powers where a prime repeats.

#	NUMBER	WORKING	PRIME FACTORISATION
a	24		
b	45		
c	72		
d	100		
e	84		
f	96		

Task 1.1b • Prime or not?

DETECTIVE

Circle every prime. Cross out the rest and write one factor pair beside it.

1 9 17 21 29 51 87 91 97 113

One of these fools almost everybody. It is odd and it is not prime. Which one, and what is it equal to?

Shared and common

KEY SKILL

18 MIN

The two questions primes can answer

GCF greatest common factor
the biggest number that
divides into **both**
Use the **lowest** power of
every **shared** prime.

LCM least common multiple
the smallest number that
both divide into
Use the **highest** power of
every prime you see.

$$18 = 2 \times 3^2$$

$$24 = 2^3 \times 3$$

$$\text{GCF} = 2 \times 3 = 6$$

$$\text{LCM} = 2^3 \times 3^2 = 72$$

Task 1.2 • Find both

DO

#	THE PAIR	PRIME FACTORS	GCF	LCM
a	12 and 18			
b	8 and 20			
c	9 and 15			
d	16 and 24			
e	14 and 21			
f	10 and 25			

Then do this. For each row multiply your GCF by your LCM, and compare it with the two starting numbers multiplied together. What do you notice?

Task 1.2b • Which one does the question want?

DECIDE

Write **GCF** or **LCM** beside each. Do not solve them yet.

#	THE SITUATION	GCF OR LCM?
a	Two ribbons, 36 cm and 48 cm, cut into equal pieces as long as possible.	
b	One alarm rings every 12 minutes, another every 18. When do they ring together?	
c	Party bags made from 24 pencils and 36 stickers with nothing left over.	
d	Two rows of tiles, 8 cm and 10 cm long, that must finish level with each other.	

What word or idea gave each one away? _____

Out in the world

18 MIN

Task 1.3 • Now solve them

APPLY

a. Ribbons of 36 cm and 48 cm are cut into equal pieces, as long as possible, with nothing wasted. How long is each piece, and how many altogether?

c. 24 pencils and 36 stickers make identical party bags with nothing left over. Most bags possible, and what is in each?

b. Bus A leaves every 12 minutes, Bus B every 18. Both leave at 9:00 am. When do they next leave together?

d. Hot dogs come in packs of 10, buns in packs of 8. Fewest packs of each so that none are left over?

Task 1.4 • Find the error

DETECTIVE

Each of these is wrong. Say what went wrong and give the correct answer.

Ravi • GCF of 12 and 30

$$12 = 2 \times 2 \times 3 \quad 30 = 2 \times 3 \times 5$$

$$\text{GCF} = 2 \times 2 \times 3 \times 5 = 60$$

What went wrong?

Correct answer: _____

Tess • LCM of 6 and 8

$$6 \times 8 = 48$$

so the LCM is 48

What went wrong?

Correct answer: _____

Owen • Is 87 prime?

87 is odd,

so yes it is prime.

What went wrong?

Correct answer: _____

Task 1.5 • Think it through

CHALLENGE

a. Two numbers have a GCF of 6 and an LCM of 36. One of them is 12. What is the other? *Use what you noticed in Task 1.2.*

b. Why can the GCF of two numbers never be larger than the smaller of the two?

c. Write two numbers whose GCF is 1. What does that tell you about them? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

The Number Detective

ANSWERS

Task 1.1 • Prime factorisation

- a. $24 = 2^3 \times 3$ d. $100 = 2^2 \times 5^2$
 b. $45 = 3^2 \times 5$ e. $84 = 2^2 \times 3 \times 7$
 c. $72 = 2^3 \times 3^2$ f. $96 = 2^5 \times 3$

f is the one to check. 96 needs five twos. A student who stops at $2^3 \times 12$ has not finished — 12 is not prime.

Task 1.1b • Prime or not

Prime: 17, 29, 97, 113.

1 → not prime (one factor only)
 $9 = 3 \times 3$ $21 = 3 \times 7$
 $51 = 3 \times 17$ $87 = 3 \times 29$
 $91 = 7 \times 13$ ← the trap

51 and 87 are also worth pointing out: adding the digits gives a multiple of 3, which is how you spot them without dividing.

Task 1.2 • GCF, LCM and the pattern

pair	GCF	LCM	GCF×LCM	pair multiplied
12, 18	6	36	216	216
8, 20	4	40	160	160
9, 15	3	45	135	135
16, 24	8	48	384	384
14, 21	7	42	294	294
10, 25	5	50	250	250

GCF × LCM always equals the two numbers multiplied together. Every row matches. A student who spots this has found something real, and it is exactly what Task 1.5a needs.

Task 1.2b • Which one

- a. GCF cutting into equal pieces
 b. LCM two cycles lining up
 c. GCF sharing out with none left
 d. LCM two lengths finishing level

The giveaway: **splitting one thing up is GCF; two things repeating until they meet is LCM.**

Task 1.3 • The word problems

- a. $\text{GCF}(36, 48) = 12$ cm each
 $3 + 4 = 7$ pieces
 b. $\text{LCM}(12, 18) = 36 \rightarrow 9:36$ am
 c. $\text{GCF}(24, 36) = 12$ bags
 2 pencils + 3 stickers in each
 d. $\text{LCM}(10, 8) = 40 \rightarrow 4$ packs of hot dogs,
 5 packs of buns

In d many students answer 40 and stop. The question asks for packs, not items.

Task 1.4 • The three errors

Ravi — GCF = 6. He used every prime he could see. The GCF takes only primes appearing in *both* lists, at the lowest power: 2×3 . What he actually worked out was the LCM.

Tess — LCM = 24. Multiplying works only when the two numbers share no factors. 6 and 8 both contain a 2, so 48 counts it twice.

Owen — $87 = 3 \times 29$. Odd does not mean prime. The digits give $8 + 7 = 15$, a multiple of 3, so 87 is too.

Task 1.5 • Challenge

- a. **18.** $\text{GCF} \times \text{LCM} =$ the two numbers multiplied, so $6 \times 36 = 216$ and $216 \div 12 = 18$. Check: $\text{GCF}(12, 18) = 6$, $\text{LCM}(12, 18) = 36$. ✓
 b. The GCF has to divide into the smaller number, and nothing bigger than a number divides into it.
 c. Any pair sharing no factor — 8 and 9, 7 and 10, 15 and 22. They are *coprime*, and for those pairs the LCM is simply the two multiplied together.

Students add, subtract, multiply and divide fractions, then use them on a recipe, a tape measure and a road trip. Dividing is the rule everybody forgets, and this lesson explains why it works rather than only stating it.

You do not need to know this subject

Every answer is a definite fraction and the key shows the working line by line. Comparing a student's answer with the printed one is the only skill you need.

Photocopy these pages only

Pages 12–14

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Calculators do not help here and are best put away.

WHAT TO SAY TO START

“Most adults cannot divide one fraction by another. You are going to be able to do it in forty minutes — and, more importantly, you will know why the rule works.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the four rules and the example	Do the common denominator on the board
14–26	Task 2.1 — the four operations	Every answer is in the key
26–38	Task 2.2 and 2.3 — recipe and amounts	Halving and scaling up
38–48	Task 2.4 — the tape measure	Eighths, the way a real ruler works
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Why do I need a common denominator to add but not to multiply?”
 Adding means counting pieces, and you can only count pieces that are the same size. Multiplying takes a part of a part, so the sizes look after themselves.

“Do I flip the first fraction or the second?”

The second one, always — the one you are dividing by. Then multiply.

“Why does dividing by a half make the number bigger?”

Ask how many halves fit inside 3. Six. Dividing asks how many fit, so dividing by something small gives a big answer.

If they finish early

Task 2.7 asks which is larger, five eighths or two thirds, without converting to decimals. Beyond that, ask them to write a division of two fractions whose answer is exactly 3.

Curriculum focus • for the regular teacher's records

Grade 6–7: adding, subtracting, multiplying and dividing fractions and mixed numbers; interpreting division of a fraction by a fraction. Discussion question for the last five minutes: *“Why does dividing by a fraction usually make the number bigger?”*

Four rules that never change

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Four rules, one page

ADD / SUBTRACT

Same denominator first.

$$\begin{aligned} \frac{1}{3} + \frac{1}{4} \\ = \frac{4}{12} + \frac{3}{12} \\ = \frac{7}{12} \end{aligned}$$

MULTIPLY

Straight across.

$$\begin{aligned} \frac{2}{3} \times \frac{3}{5} \\ = \frac{6}{15} \\ = \frac{2}{5} \end{aligned}$$

DIVIDE

Flip the second, multiply.

$$\begin{aligned} \frac{3}{4} \div \frac{1}{2} \\ = \frac{3}{4} \times \frac{2}{1} = \frac{6}{4} \\ = 1 \frac{1}{2} \end{aligned}$$

Worked example • $\frac{2}{3} + \frac{1}{4}$ denominators 3 and 4 → common denominator **12**

$$\frac{2}{3} = \frac{8}{12} \quad \frac{1}{4} = \frac{3}{12}$$

$$\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$$

The denominator never gets added. You are counting twelfths, and 8 twelfths plus 3 twelfths is 11 twelfths.

Task 2.1 • Add and subtract

DO

Show the common denominator step. Simplify your answer where you can.

#	PROBLEM	WORKING	ANSWER	#	PROBLEM	WORKING	ANSWER
a	$\frac{1}{2} + \frac{1}{3}$			d	$\frac{5}{6} - \frac{1}{3}$		
b	$\frac{3}{4} - \frac{1}{2}$			e	$\frac{2}{3} + \frac{3}{4}$		
c	$\frac{2}{5} + \frac{3}{10}$			f	$\frac{7}{8} - \frac{1}{6}$		

Task 2.1b • Multiply and divide

DO

#	PROBLEM	WORKING	ANSWER	#	PROBLEM	WORKING	ANSWER
g	$\frac{1}{2} \times \frac{2}{3}$			j	$\frac{2}{3} \div 4$		
h	$\frac{3}{5} \times 10$			k	$\frac{5}{6} \times \frac{3}{10}$		
i	$\frac{3}{4} \div \frac{1}{2}$			l	$\frac{1}{2} \div \frac{1}{8}$		

In i, j and l you divided. Two answers got bigger and one got smaller. What made the difference?

Where fractions actually turn up

KEY SKILL

20 MIN

Task 2.2 • The recipe

SCALE

This recipe serves **6**. Fill in the amounts for 3 servings (half it) and for 9 servings (one and a half times).

INGREDIENT	SERVES 6	SERVES 3	SERVES 9
Flour (cups)	2 1/4		
Sugar (cups)	3/4		
Oil (cups)	2/3		
Butter (oz)	8		
Milk (cups)	1 1/2		
Salt (tsp)	1/2		

Which column did you find harder, and why? _____

Task 2.3 • Fraction of an amount

DRILL

“Of” means multiply. There is no need to write the whole number as a fraction first — divide by the bottom, multiply by the top.

FIND	ANSWER	FIND	ANSWER
2/3 of 45		5/6 of 42	
3/4 of 60		2/5 of 35	
3/8 of 32		7/8 of 24	

Task 2.4 • The tape measure

APPLY

Real rulers run in eighths. Give every answer as a mixed number in its simplest form.

a. A board is $7\frac{7}{8}$ in long. Two of them are laid end to end. How long is that?

c. How many $\frac{3}{4}$ in pieces can be cut from a board 12 in long?

b. $\frac{3}{8}$ in is cut off one board. How much of the $7\frac{7}{8}$ in board is left?

d. Three shelves, each $5\frac{1}{4}$ in wide, sit side by side. Total width?

Fractions that decide things

STRETCH

17 MIN

Task 2.5 • The road trip

APPLY

A trip is **240 miles** long.

- They drive $\frac{3}{8}$ of it before lunch. How far is that? _____
- How far is left after lunch? _____
- After lunch they cover $\frac{2}{3}$ of what remained. How far is that? _____
- How far still to go? _____
- What fraction of the *whole* trip is still left? _____

Task 2.6 • Find the error

DETECTIVE

Each of these is wrong in a way that is very common. Say what happened.

Priya

$$\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$$

What went wrong?

Correct answer: _____

Sam

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

What went wrong?

Correct answer: _____

Rey

$$\frac{2}{3} \times 6 = \frac{2}{18}$$

What went wrong?

Correct answer: _____

Task 2.7 • Think it through

CHALLENGE

- Which is larger, $\frac{5}{8}$ or $\frac{2}{3}$? Show how you know without turning either into a decimal.

- Half a pizza is left. You eat $\frac{2}{3}$ of what is left. What fraction of the *whole* pizza did you eat?

- Write a division of two fractions whose answer is exactly 3. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Slices and Shares

ANSWERS

Task 2.1 • Add and subtract

- a. $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$
 b. $\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$
 c. $\frac{4}{10} + \frac{3}{10} = \frac{7}{10}$
 d. $\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$
 e. $\frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1 \frac{5}{12}$
 f. $\frac{21}{24} - \frac{4}{24} = \frac{17}{24}$

e is the argument starter — the answer is bigger than 1, which surprises students who expect fractions to stay small.

Task 2.1b • Multiply and divide

- g. $\frac{2}{6} = \frac{1}{3}$
 h. $\frac{30}{5} = 6$
 i. $\frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = 1 \frac{1}{2}$
 j. $\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$
 k. $\frac{15}{60} = \frac{1}{4}$
 l. $\frac{1}{2} \times \frac{8}{1} = \frac{8}{2} = 4$

i and l divided by something *smaller* than 1, so the answer grew. j divided by 4, which is bigger than 1, so it shrank. That is the whole idea.

Task 2.2 • The recipe

	serves 6	serves 3	serves 9
Flour	2 $\frac{1}{4}$	1 $\frac{1}{8}$	3 $\frac{3}{8}$
Sugar	$\frac{3}{4}$	$\frac{3}{8}$	1 $\frac{1}{8}$
Oil	$\frac{2}{3}$	$\frac{1}{3}$	1
Butter	8	4	12
Milk	1 $\frac{1}{2}$	$\frac{3}{4}$	2 $\frac{1}{4}$
Salt	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$

Serves 9 is the harder column because it means multiplying by 1 $\frac{1}{2}$, not doubling. Accept 1.5 $\times$ as working.

Task 2.3 • Fraction of an amount

- $\frac{2}{3}$ of 45 = **30** $\frac{5}{6}$ of 42 = **35**
 $\frac{3}{4}$ of 60 = **45** $\frac{2}{5}$ of 35 = **14**
 $\frac{3}{8}$ of 32 = **12** $\frac{7}{8}$ of 24 = **21**

Divide by the bottom, multiply by the top. A student doing $45 \div 3 = 15$ then $15 \times 2 = 30$ has it exactly right.

Task 2.4 • The tape measure

- a. $7 \frac{7}{8} + 7 \frac{7}{8} = 15 \frac{3}{4}$
 b. $7 \frac{7}{8} - 3 \frac{7}{8} = 4$
 c. $12 \div \frac{3}{4} = 16$ pieces
 d. $5 \frac{1}{4} \times 3 = 15 \frac{3}{4}$

c is the one worth stopping on: dividing gave a bigger number than 12, which is correct.

Task 2.5 • The road trip

- a. $\frac{3}{8}$ of 240 = **90 miles**
 b. $240 - 90 = 150$ miles
 c. $\frac{2}{3}$ of 150 = **100 miles**
 d. $150 - 100 = 50$ miles
 e. $\frac{50}{240} = \frac{5}{24}$

In c, students who take $\frac{2}{3}$ of 240 instead of $\frac{2}{3}$ of the *remainder* get 160. Point at the word “remained”.

Task 2.6 • The three errors

Priya — $\frac{7}{12}$. She added the denominators. Thirds and quarters are different-sized pieces and cannot be counted together until both are twelfths.

Sam — 2. He multiplied instead of flipping. Ask how many quarters fit into a half: two.

Rey — 4. He multiplied the bottom by 6 as well. 6 is $\frac{6}{1}$, so $\frac{2}{3} \times \frac{6}{1} = \frac{12}{3} = 4$.

Task 2.7 • Challenge

a. $\frac{2}{3}$ is larger. Over 24: $\frac{5}{8} = \frac{15}{24}$ and $\frac{2}{3} = \frac{16}{24}$. One twenty-fourth in it.

b. $\frac{1}{3}$ of the whole pizza. $\frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$. Students who answer $\frac{2}{3}$ have forgotten it was only half a pizza to begin with.

c. Any correct pair — $\frac{3}{4} \div \frac{1}{4}$, $\frac{1}{2} \div \frac{1}{6}$, $\frac{9}{10} \div \frac{3}{10}$. Have them check by multiplying the answer back.

Scale It Up

2 MINUTES

Students test whether a table is proportional, pull out the one number that controls it, then use that number on a floor plan, a map and a model car. Ends with the fact that surprises everybody: doubling the lengths does not double the area.

You do not need to know this subject

Every answer is a multiplication or a division and the key shows both. The only judgment call is whether a table is proportional, and the key explains exactly how to check.

Photocopy these pages only

Pages 17–19

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler if you have one. Calculators are fine and realistic here.

WHAT TO SAY TO START

“A model car, a map and a set of blueprints all do the same job. They shrink something real by one fixed amount and keep every part in the same relationship. Today you find that one number and use it.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the panel and worked example	Divide y by x for every row
14–26	Task 3.1 — is it proportional?	One table fails only in the last row
26–40	Task 3.2 — the floor plan	Drawing to real, multiply
40–50	Task 3.3 — the map	Real to drawing, divide
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“How do I know if it is proportional?”

Divide y by x for every row. If the answer is identical every time, it is. One row that disagrees fails the whole table.

“Which number goes on top?”

Whatever the question wants per one. Miles per hour puts miles on top.

“The scale says 1 cm : 2 m. Do I multiply or divide?”

Going from the drawing to the real thing, multiply. Going from real life onto the paper, divide.

If they finish early

Task 3.6 asks about a model car where the area does not scale the way students expect. Beyond that, ask them what scale would be needed to fit the whole classroom on one page.

Curriculum focus • for the regular teacher’s records

Grade 7: proportional relationships, the constant of proportionality, and scale drawings of geometric figures. Discussion question for the last five minutes: *“A map without its scale printed on it is nearly useless. Why?”*

Find the one number

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

One number runs the whole table

A relationship is **proportional** if $y \div x$ gives the same answer every single time. That answer is **k**, and it turns the table into a rule.

$$y = k \times x$$

k is the amount of y
for exactly one x.

Every row has to agree. One row that does not sinks the whole table.

hours	2	5	8
pay	\$34	\$85	\$136

$$34 \div 2 = 17 \quad 85 \div 5 = 17 \quad 136 \div 8 = 17$$

k = 17 → pay = 17 × hours
20 hours → **\$340**

Task 3.1 • Is it proportional?

TEST

Divide y by x for every column. If k is the same each time, write it and the equation. If not, write **not proportional**.

x	1	2	3	4
y	3	6	9	12

A k = _____ y = _____

x	1	2	3	4
y	5	7	9	11

B k = _____ y = _____

x	2	4	6	8
y	5	10	15	20

C k = _____ y = _____

x	1	2	3	4
y	4	8	12	15

D k = _____ y = _____

One table works for three rows and fails on the fourth. Which, and why does one bad row sink it? _____

Task 3.1b • Fill in the missing value

DO

Each of these is proportional. Find k first.

#	THE SITUATION	K	ANSWER
a	3 pens cost \$4.50. Cost of 7 pens?		
b	150 miles in 3 hours. How far in 7 hours?		
c	4 kg of apples cost \$9.00. Cost of 10 kg?		
d	24 pages in 3 minutes. How many in 11 min?		

Plans and rates

KEY SKILL

18 MIN

What a scale actually says

1 cm : 2 mone centimetre on paper
means two metres in real life**drawing → real** multiply

$$4 \text{ cm} \times 2 = 8 \text{ m}$$

real → drawing divide

$$12 \text{ m} \div 2 = 6 \text{ cm}$$

Task 3.2 • The floor plan

APPLY

This plan is drawn at **1 cm : 2 m**. Fill in every gap.

ROOM	ON THE PLAN	WORKING	REAL LENGTH
Kitchen	4 cm		
Bedroom	5.5 cm		
Hallway	3.5 cm		
Garage			12 m
Bathroom	1.5 cm		

The kitchen measures **4 cm × 3 cm** on the plan. Its real area is _____ m², and its area on the plan is _____ cm².

The lengths were multiplied by 2. What happened to the area, and why?

Task 3.4 • Who is fastest?

COMPARE

Three runners, three different distances. You cannot compare them until every one is a rate.

RUNNER	DISTANCE	TIME	WORKING	SPEED
Ana	12 km	1.5 h		
Ben	21 km	3 h		
Cara	9 km	1 h 15 min		

Fastest: _____ 20 km at that speed would take _____

Maps, models and one surprise

STRETCH

20 MIN

Task 3.3 • The map

APPLY

A road map is printed at **1 inch : 25 miles**.

a. Two towns are 3.5 in apart on the map. How far apart are they really?

c. A road measures $6\frac{1}{4}$ in on the map. How long is it?

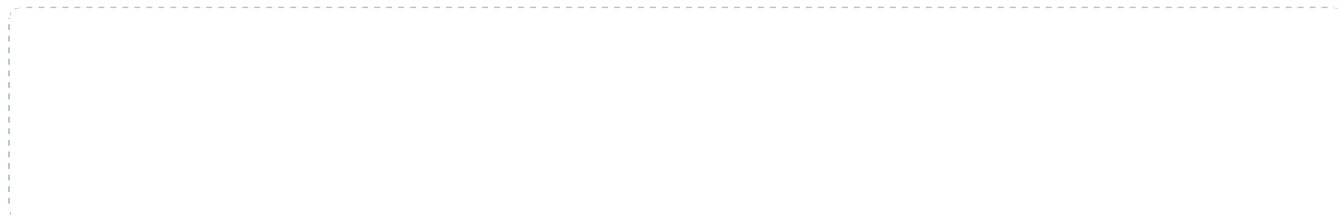
b. Two cities are 200 miles apart. How far apart on the map?

d. Driving at 50 mph, how long would the trip in part a take?

Task 3.5 • Draw it to scale

CREATE

A door is **200 cm** tall and **80 cm** wide. Draw it at **1 cm : 20 cm** and label both sides.



On your drawing the door is _____ cm tall and _____ cm wide.

The whole classroom is 8 m by 6 m. What scale would fit it on this page? _____

Task 3.6 • Think it through

CHALLENGE

A model car is built at a scale of **1 : 24**.

a. The model is 7.5 cm long. Real car? _____ b. A real wheel is 60 cm across. Model wheel? _____

c. The real windscreen has an area of $5,760\text{ cm}^2$. A student says the model's is $5,760 \div 24 = 240\text{ cm}^2$. What did they get wrong, and what is the right answer?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Scale It Up

ANSWERS

Task 3.1 • Is it proportional?

- | | |
|-------------------|-------------------------|
| A 3, 3, 3, 3 | $k = 3$ $y = 3x$ |
| B 5, 3.5, 3, 2.75 | not proportional |
| C 2.5 every time | $k = 2.5$ $y = 2.5x$ |
| D 4, 4, 4, 3.75 | not proportional |

D is the trap. Three rows agree and the fourth does not, so the rule does not hold everywhere — and a rule with an exception is no rule. B rises steadily by 2 each time, which looks convincing until you divide.

Task 3.1b • Missing values

- | | | |
|---------------|--------------------|------------------|
| a. $k = 1.50$ | $7 \times 1.50 =$ | \$10.50 |
| b. $k = 50$ | $7 \times 50 =$ | 350 miles |
| c. $k = 2.25$ | $10 \times 2.25 =$ | \$22.50 |
| d. $k = 8$ | $11 \times 8 =$ | 88 pages |

Finding k first is slower on one question and far faster on five. Say so.

Task 3.2 • The floor plan

- | | | | |
|----------|-------------|------------------------|-------------|
| Kitchen | 4 cm | $\times 2 =$ | 8 m |
| Bedroom | 5.5 cm | $\times 2 =$ | 11 m |
| Hallway | 3.5 cm | $\times 2 =$ | 7 m |
| Garage | 6 cm | $\leftarrow 12 \div 2$ | |
| Bathroom | 1.5 cm | $\times 2 =$ | 3 m |

Kitchen: plan area **12 cm²**, real area $8 \times 6 =$ **48 m²**. The lengths doubled but the area went up **four** times, because area is made of two lengths and both of them doubled. This is the single most valuable idea on the page and it comes back in Task 3.6.

Task 3.3 • The map

- | | | |
|---------------------|-----|----------------------------|
| a. 3.5×25 | $=$ | 87.5 miles |
| b. $200 \div 25$ | $=$ | 8 inches |
| c. 6.25×25 | $=$ | 156.25 miles |
| d. $87.5 \div 50$ | $=$ | 1.75 h = 1 h 45 min |

b is the only one that divides. Students who multiply get 5,000 inches — worth asking whether that map would fit in a car.

Task 3.4 • Who is fastest?

- | | | | | |
|------|---------------|-----|-----------------|---------|
| Ana | $12 \div 1.5$ | $=$ | 8 km/h | fastest |
| Ben | $21 \div 3$ | $=$ | 7 km/h | |
| Cara | $9 \div 1.25$ | $=$ | 7.2 km/h | |

20 km at 8 km/h = **2.5 hours**

Cara is the one to watch — 1 h 15 min has to become 1.25 h, not 1.15.

Task 3.5 • The door

- | | | |
|---------------|-----|-------------------|
| $200 \div 20$ | $=$ | 10 cm tall |
| $80 \div 20$ | $=$ | 4 cm wide |

For the classroom, 8 m by 6 m is 800 cm by 600 cm. At 1 cm : 50 cm that is 16 cm by 12 cm, which fits. Accept any scale that produces a drawing smaller than the page — there are many right answers and the reasoning is what counts.

Task 3.6 • The model car

a. $7.5 \times 24 =$ **180 cm**, or 1.8 m. b. $60 \div 24 =$ **2.5 cm**.

c. **The area is 10 cm², not 240.** Dividing by 24 once shrinks a single length. An area is made of two lengths, so it shrinks by $24 \times 24 = 576$. $5,760 \div 576 = 10$ cm². Any student who connects this back to the kitchen in Task 3.2 has understood the most important idea in the lesson.

Powers of Ten

2 MINUTES

Students evaluate powers, work out the multiply and divide rules by looking at what an exponent actually counts, then write real measurements in scientific notation — the width of a hair, the distance to the moon, the speed of light.

You do not need to know this subject

Every answer is a single number or a single power, and the key prints all of them. Nothing here requires you to remember an exponent rule.

Photocopy these pages only

Pages 22–24

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Calculators are not needed and mostly get in the way.

WHAT TO SAY TO START

“A bacterium is 0.000002 metres across. The sun is 1,400,000,000 metres across. Writing those out is a nuisance and it is far too easy to lose a zero. Today you learn the shorthand scientists actually use.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the powers panel	Stress that 3^2 is not 3×2
12–24	Task 4.1 — evaluate and use the rules	Straight practice
24–36	Task 4.2 — scientific notation	Which way does the point move?
36–44	Task 4.3 — real measurements	The numbers here are genuine
44–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is 3^2 the same as 3×2 ?”

No. $3^2 = 3 \times 3 = 9$. This is the single most common mistake in the lesson and it is worth saying out loud before they start.

“Why is anything to the power zero equal to 1?”

Look at $2^3 = 8$, $2^2 = 4$, $2^1 = 2$. Each step halves. The next step gives 1.

“Which way does the decimal point move?”

A positive power makes the number bigger, so the point goes right. A negative power makes it smaller, so the point goes left.

If they finish early

Task 4.6 asks how thick a sheet of paper would be after twenty folds. The answer is over a hundred metres and nobody believes it until they check.

Curriculum focus • for the regular teacher’s records

Grade 8: integer exponents and their properties; expressing and comparing numbers in scientific notation. Discussion question for the last five minutes:

“Why is 4.5×10^8 easier to work with than 450,000,000?”

What an exponent means

UNDERSTAND

14 MIN

NAME _____

CLASS _____

DATE _____

An exponent counts how many, not how much

$$5^3 = 5 \times 5 \times 5 = 125$$

base **5**, exponent **3**

5^3 is **not** 5×3

$$10^2 \times 10^3 = 10^5 \quad \text{add them}$$

$$10^6 \div 10^2 = 10^4 \quad \text{subtract}$$

$$(10^2)^3 = 10^6 \quad \text{multiply}$$

$$10^0 = 1 \quad \text{always}$$

Why adding works:

$$\begin{aligned} 10^2 \times 10^3 &= (10 \times 10) \times (10 \times 10 \times 10) \\ &= \text{five tens in a row} \\ &= 10^5 \end{aligned}$$

Task 4.1 • Work them out

DO

#	POWER	WORKING	ANSWER	#	POWER	WORKING	ANSWER
a	2^4			e	2^6		
b	3^3			f	4^3		
c	5^2			g	1^9		
d	10^4			h	7^0		

Two of these give the same answer. Which two, and does that mean 2 and 4 are somehow the same number?

Task 4.1b • Use the rules

DO

Leave every answer as a power. Do not work out the number.

#	EXPRESSION	ANSWER	#	EXPRESSION	ANSWER
a	$10^3 \times 10^4$		d	$(10^3)^2$	
b	$10^8 \div 10^3$		e	$5^6 \div 5^6$	
c	$2^5 \times 2^2$		f	$3^2 \times 3^3 \div 3^4$	

Very big and very small

KEY SKILL

18 MIN

The shorthand

$$a \times 10^n$$

a is at least 1
and less than 10.

$4,500 = 4.5 \times 10^3$
point moved **3 left**
big number → **positive**

$0.0072 = 7.2 \times 10^{-3}$
point moved **3 right**
small number → **negative**

Task 4.2 • Into scientific notation

DO

#	NUMBER	SCIENTIFIC NOTATION	#	NUMBER	SCIENTIFIC NOTATION
a	3,200		d	0.005	
b	47,000		e	0.00082	
c	6,000,000		f	125,000,000	

Task 4.2b • And back out again

DO

#	SCIENTIFIC NOTATION	ORDINARY NUMBER	#	SCIENTIFIC NOTATION	ORDINARY NUMBER
a	5×10^4		d	3×10^{-5}	
b	2.6×10^3		e	7.04×10^5	
c	9.1×10^{-2}		f	1.5×10^{-1}	

Task 4.3 • Real measurements

APPLY

These are the actual figures. Fill in the empty column.

WHAT IT IS	ORDINARY NUMBER	SCIENTIFIC NOTATION
Distance to the moon	384,000 km	
Width of a human hair	0.00007 m	
People on Earth	8,000,000,000	
Mass of a grain of sand	0.000011 kg	
Speed of light	300,000,000 m/s	

Compare, correct, and fold

STRETCH

18 MIN

Task 4.4 • Which is bigger?

COMPARE

Circle the larger one. Do not convert them to ordinary numbers — you do not need to.

#	FIRST	SECOND	WHY
a	3×10^5	9×10^4	
b	6.1×10^{-3}	5.9×10^{-2}	
c	2×10^8	1.9×10^8	

What do you compare first, and what only matters if that is a tie? _____

Task 4.5 • Find the error

DETECTIVE

Jae

$$4^3 = 12$$

What went wrong?

Correct answer: _____

Mia

$$\begin{aligned} 0.0043 \\ = 4.3 \times 10^3 \end{aligned}$$

What went wrong?

Correct answer: _____

Tom

$$\begin{aligned} 10^2 + 10^3 \\ = 10^5 \end{aligned}$$

What went wrong?

Correct answer: _____

Task 4.6 • Think it through

CHALLENGE

a. A sheet of paper is 0.1 mm thick. Every fold doubles it. Twenty folds multiplies the thickness by 2^{20} , which is 1,048,576. How thick is it, in metres?

b. How many times bigger is 6×10^8 than 3×10^5 ? _____

c. Use the pattern $2^3 = 8$, $2^2 = 4$, $2^1 = 2$ to explain why 2^0 must be 1.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Powers of Ten

ANSWERS

Task 4.1 • Work them out

- a. $2^4 = 16$ e. $2^6 = 64$
 b. $3^3 = 27$ f. $4^3 = 64$
 c. $5^2 = 25$ g. $1^9 = 1$
 d. $10^4 = 10,000$ h. $7^0 = 1$

e and f both give 64 — not because 2 and 4 are the same, but because 4 is 2^2 , so 4^3 is really 2^6 in disguise. Award credit generously here; spotting it is genuinely sharp.

Task 4.1b • Use the rules

- a. $10^3 \times 10^4 = 10^7$
 b. $10^8 \div 10^3 = 10^5$
 c. $2^5 \times 2^2 = 2^7$
 d. $(10^3)^2 = 10^6$
 e. $5^6 \div 5^6 = 5^0 = 1$
 f. $3^2 \times 3^3 \div 3^4 = 3^{5-4} = 3$

e is the one that explains the zero rule: any number divided by itself is 1.

Task 4.2 • Into scientific notation

- a. 3,200 = 3.2×10^3
 b. 47,000 = 4.7×10^4
 c. 6,000,000 = 6×10^6
 d. 0.005 = 5×10^{-3}
 e. 0.00082 = 8.2×10^{-4}
 f. 125,000,000 = 1.25×10^8

Task 4.2b • And back

- a. $5 \times 10^4 = 50,000$
 b. $2.6 \times 10^3 = 2,600$
 c. $9.1 \times 10^{-2} = 0.091$
 d. $3 \times 10^{-5} = 0.00003$
 e. $7.04 \times 10^5 = 704,000$
 f. $1.5 \times 10^{-1} = 0.15$

Task 4.3 • Real measurements

Moon	3.84×10^5 km
Hair	7×10^{-5} m
People	8×10^9
Sand	1.1×10^{-5} kg
Light	3×10^8 m/s

Task 4.4 • Which is bigger

- a. 3×10^5 — 300,000 against 90,000. A bigger power wins even though 3 is smaller than 9.
 b. 5.9×10^{-2} — 0.059 against 0.0061. With negatives, -2 is the larger power.
 c. 2×10^8 — the powers tie, so the front number decides.

The rule: compare the power first. The leading number only matters if the powers are equal.

Task 4.5 • The three errors

- Jae — 64.** He multiplied the base by the exponent. 4^3 means $4 \times 4 \times 4$.
Mia — 4.3×10^{-3} . 0.0043 is smaller than 1, so the power has to be negative. A positive power would make it 4,300.
Tom — 1.1×10^3 . The add-the-exponents rule is for multiplying, not adding. $100 + 1,000 = 1,100$.

Task 4.6 • Challenge

- a. **About 105 metres.** $0.1 \text{ mm} \times 1,048,576 = 104,857.6 \text{ mm}$, which is 104.86 m — taller than a thirty-storey building, from twenty folds of one sheet. Accept anything close, and accept 1.05×10^2 m.
 b. **2,000 times.** $6 \div 3 = 2$ and $10^8 \div 10^5 = 10^3$, so 2×10^3 .
 c. Each step down divides by 2: 8, 4, 2 ... so the next one is 1. The pattern forces it. Accept any answer that says the pattern would break otherwise.

Same Thing, Shorter

READ BEFORE THE BELL

2 MINUTES

Students collect like terms, substitute numbers into expressions, expand brackets and factor them back again. It finishes with two expressions that look nothing alike and are in fact identical — which students can test for themselves.

You do not need to know this subject

Every answer is printed in the key, and every one can be checked by putting a number in. If both expressions give the same value, the student is right.

Photocopy these pages only

Pages 27–29

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Calculators are not needed.

WHAT TO SAY TO START

" $3x + 5x$ is $8x$, but $3x + 5$ is not $8x$, and knowing the difference is most of algebra. Today you get very fast at telling which terms can be joined and which cannot."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the like terms panel	Three boxes plus five apples
14–26	Task 5.1 — collect and substitute	Watch the signs
26–40	Task 5.2 — expanding brackets	Multiply <i>everything</i> inside
40–50	Task 5.3 and 5.4 — factor, then words	Factoring is expanding backwards
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Why can't I add $3x$ and 5 ?"

Three boxes plus five apples is not eight of anything. Terms only join when the letter part is identical.

"Is $2x$ the same as x^2 ?"

No. $2x$ is $x + x$. x^2 is $x \times x$. Put $x = 5$ in and one gives 10, the other 25.

"Do I multiply the outside number by everything inside?"

Yes, every single term. Missing the second one is the most common error on the page and Task 5.5 has an example of it.

If they finish early

Task 5.6 asks them to show that two different-looking expressions are the same and then test both with a number. Beyond that, ask for two more expressions that both equal $12x + 18$.

Curriculum focus • for the regular teacher's records

Grade 6–7: generating equivalent expressions, combining like terms, applying the distributive property and evaluating expressions at given values. Discussion question for the last five minutes: *"Why is $3x + 5x$ allowed but $3x + 5$ not?"*

Which terms can join

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Terms only join when they match

$3x^2 - 5x + 7$

three terms

The number in front is the **coefficient**.

The sign travels with the term.

LIKE TERMS

same letter, same power

3x and 5x **yes**3x and $3x^2$ **no**3x and 5 **no**2xy and 7xy **yes**

$4x + 7 + 2x - 3$

$= 4x + 2x + 7 - 3$

$= 6x + 4$

Group first, then combine. $6x + 4$ will not go any further.

Task 5.1 • Collect the like terms

DO

Some of these are already as short as they go. Say so if they are.

#	EXPRESSION	SIMPLIFIED
a	$5x + 3x$	
b	$9y - 4y$	
c	$2a + 7 + 3a$	
d	$6m + 4 - 2m - 9$	

#	EXPRESSION	SIMPLIFIED
e	$3x + 2y + 5x - y$	
f	$8 - 3k + 2k + 1$	
g	$4p + p$	
h	$7c - 7c + 3$	

Task 5.1b • Put the numbers in

DO

Throughout this task $x = 4$ and $y = 3$.

#	EXPRESSION	VALUE
a	$2x + 5$	
b	$5y - 7$	
c	xy	
d	x^2	

#	EXPRESSION	VALUE
e	$2x^2$	
f	$(2x)^2$	
g	$3x + 2y$	
h	$x/2 + y$	

Look at e and f. Why do they give different answers? _____

Opening and closing brackets

KEY SKILL

20 MIN

Brackets, both ways

EXPAND multiply everything inside

$$3(x + 4) = 3x + 12$$

$$5(2a - 3) = 10a - 15$$

$$-2(x + 6) = -2x - 12$$

FACTOR pull the common number out

$$6x + 15 = 3(2x + 5)$$

Check by expanding it back.
They undo each other.

Task 5.2 • Expand

DO

#	EXPRESSION	EXPANDED	#	EXPRESSION	EXPANDED
a	$4(x + 3)$		d	$-3(x + 4)$	
b	$2(3y - 5)$		e	$5(2m - 3n)$	
c	$6(2a + 1)$		f	$x(x + 7)$	

Task 5.2b • Expand, then simplify

DO

Two steps. Expand every bracket first, then collect what is left.

#	EXPRESSION	EXPANDED	SIMPLIFIED
a	$2(x + 3) + 4x$		
b	$3(a + 2) + 5(a - 1)$		
c	$4(2y + 1) - 3y$		
d	$5(x - 2) - 2(x + 3)$		

Task 5.3 • Factor it back

REVERSE

Pull out the biggest number that divides into both terms.

#	EXPRESSION	FACTORED	#	EXPRESSION	FACTORED
a	$4x + 12$		c	$6y + 9$	
b	$10a - 15$		d	$8m - 20$	

Words, errors and equivalence

STRETCH

19 MIN

Task 5.4 • From words to expressions

APPLY

Let x be the unknown. Simplify wherever you can.

#	IN WORDS	EXPRESSION
a	Three times a number, minus 7	
b	A number doubled, then 4 added, then all of it tripled	
c	Ana has n dollars. Ben has \$6 more. Write the total.	
d	A taxi charges \$4 plus \$2.50 a mile. Write the cost of m miles.	
e	A rectangle is x cm wide and 3 cm longer than that. Write its perimeter.	

Task 5.5 • Find the error

DETECTIVE

Zoe

$$3x + 4$$

$$= 7x$$

What went wrong?

Correct answer: _____

Kian

$$2(x + 5)$$

$$= 2x + 5$$

What went wrong?

Correct answer: _____

Mira

$$6x - 2x$$

$$= 4$$

What went wrong?

Correct answer: _____

Leo

$$-3(x - 2)$$

$$= -3x - 6$$

What went wrong?

Correct answer: _____

Task 5.6 • Think it through

CHALLENGE

a. Show that $2(3x + 4) + 5$ and $6x + 13$ are the same expression. Then put $x = 10$ into both and check.

b. Write two different-looking expressions that both equal $12x + 18$. _____

c. Is $4(x + 2)$ ever equal to $4x + 2$? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Same Thing, Shorter

ANSWERS

Task 5.1 • Collect like terms

- | | |
|-------------|-------------|
| a. $8x$ | e. $8x + y$ |
| b. $5y$ | f. $9 - k$ |
| c. $5a + 7$ | g. $5p$ |
| d. $4m - 5$ | h. 3 |

c and d cannot go further — $5a$ and 7 are not like terms. Accept $-k + 9$ for f, and note that in h the x-terms cancel completely, which is allowed.

Task 5.1b • Substitution ($x = 4$, $y = 3$)

- | | |
|------------------|------|
| a. $2(4) + 5$ | = 13 |
| b. $5(3) - 7$ | = 8 |
| c. 4×3 | = 12 |
| d. 4^2 | = 16 |
| e. 2×16 | = 32 |
| f. 8^2 | = 64 |
| g. $12 + 6$ | = 18 |
| h. $2 + 3$ | = 5 |

e and f are the point of the task. In e you square first and then double. In f you double first and then square. Order matters, and the brackets are what say so.

Task 5.2 • Expand

- | | |
|--------------|----------------|
| a. $4x + 12$ | d. $-3x - 12$ |
| b. $6y - 10$ | e. $10m - 15n$ |
| c. $12a + 6$ | f. $x^2 + 7x$ |

d is the sign trap: -3 multiplies the 4 as well, so the second term is also negative.

Task 5.2b • Expand then simplify

- | | | |
|-----------------------|---|-----------|
| a. $2x + 6 + 4x$ | = | $6x + 6$ |
| b. $3a + 6 + 5a - 5$ | = | $8a + 1$ |
| c. $8y + 4 - 3y$ | = | $5y + 4$ |
| d. $5x - 10 - 2x - 6$ | = | $3x - 16$ |

In d the -2 has to hit both the x and the $+3$. Students who write $-2x + 6$ get $3x - 4$ and should be sent back to that line.

Task 5.3 • Factor

- | |
|----------------|
| a. $4(x + 3)$ |
| b. $5(2a - 3)$ |
| c. $3(2y + 3)$ |
| d. $4(2m - 5)$ |

Check any answer by expanding it. If it does not come back to the original, it is wrong — and the student can see that themselves.

Task 5.4 • From words

- | | |
|---------------------------|-----------------------------|
| a. $3x - 7$ | d. $4 + 2.5m$ |
| b. $3(2x + 4) = 6x + 12$ | e. $2x + 2(x + 3) = 4x + 6$ |
| c. $n + (n + 6) = 2n + 6$ | |

e is the hard one. The rectangle is x by $x + 3$, so the perimeter is two of each. Accept the unsimplified form as partly right.

Task 5.5 • The four errors

Zoe — $3x + 4$ does not simplify. They are not like terms.

Kian — $2x + 10$. He multiplied the x but not the 5.

Mira — $4x$. She dropped the letter. Six x take away two x leaves four of them, not four.

Leo — $-3x + 6$. Negative times negative is positive.

Kian's and Leo's matter most — both come from bracket expansion, where most of next year's algebra errors start.

Task 5.6 • Challenge

a. $2(3x + 4) + 5 = 6x + 8 + 5 = 6x + 13$. At $x = 10$ both give 73. Insist on the check — it is the habit worth taking away.

b. $6(2x + 3)$, $3(4x + 6)$, $2(6x + 9)$ and $12x + 18$ itself all work. Any answer that expands back is correct.

c. **Never.** $4(x + 2) = 4x + 8$, and 8 is never equal to 2, whatever x is. Accept a student who tests two values and says it fails both times.

More Than, Less Than

READ BEFORE THE BELL

2 MINUTES

Students read inequality symbols, mark solutions on a number line, solve one- and two-step inequalities, and meet the one move that catches everybody — dividing by a negative and flipping the sign. Finishes on real constraints: budgets, lift capacities and a grade average.

You do not need to know this subject

Every answer is in the key, and every one can be tested by putting a number in. If the number makes the statement true, it belongs in the answer.

Photocopy these pages only

Pages 32–34

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler if you have one. Calculators are not needed.

WHAT TO SAY TO START

"An equation has one answer. An inequality usually has millions, and you have to describe them all at once. Today you learn how — and you find the one move that turns everything upside down."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the symbols panel	Open circle or filled?
14–26	Task 6.1 — write it in symbols	'At least' and 'at most' include the number
26–38	Tasks 6.2 to 6.4 — graph, solve, correct	One step each
38–50	Tasks 6.5 and 6.6 — two steps and real limits	Two of these flip. Let them find out
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Open circle or filled in?"

Filled if the number itself counts ($\leq$ or $\geq$). Open if it does not ($<$ or $>$).

"Which way does the arrow point?"

Toward the numbers that work. Pick one and test it — that settles it every time.

"Why does dividing by a negative flip the sign?"

$3 < 5$ is true. Multiply both sides by -1 and you get -3 and -5 — but now -3 is the *bigger* one. The order reverses. Point at a number line.

If they finish early

Task 6.7 asks for an inequality that no number satisfies, and one that every number satisfies. Both exist and both surprise students.

Curriculum focus • for the regular teacher's records

Grade 6–7: writing, solving and graphing one- and two-step inequalities, and interpreting solutions in context. Discussion question for the last five minutes: "Why does an equation have one answer but an inequality has millions?"

Describing millions of answers

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Four symbols and one circle

- $<$ less than
 $>$ greater than
 $\leq$ less than or equal
 $\geq$ greater than or equal

- $x > 3$ every number above 3
 but not 3 itself
 $x \geq 3$ above 3, and 3 too

 $x > 2$ — open circle, 2 does not count $x \leq -1$ — filled circle, -1 counts

Task 6.1 • Write it in symbols

DO

Use the letter given. You are not solving anything yet.

#	IN WORDS	LETTER	INEQUALITY
a	You must be at least 13 to open the account.	a	
b	The lift holds no more than 8 people.	p	
c	She spent less than \$20.	s	
d	The temperature stayed above 5 degrees.	t	
e	You need more than 60 points to pass.	n	
f	The bag can weigh 23 kg at most.	w	

Two of these say “at least” or “at most”. Why do those get the line under the symbol? _____

Task 6.1b • Does it work?

TEST

Circle every value that makes the statement true.

 $x \leq 4$ 4 4.5 0 -7 10 3.99 $x > -2$ 0 -2 -1.9 -5 100 -2.1

Task 6.2 • Show it on the line

PLOT

Mark the circle, fill it or leave it open, then draw the arrow.

a. $x > 1$



c. $x \geq 0$



b. $x \leq -3$



d. $x < 5$



Solve it like an equation, with one exception

**IF YOU MULTIPLY OR DIVIDE
BY A NEGATIVE, FLIP THE SYMBOL.**

Adding and subtracting never flip anything.

$$-2x < 8$$

$$-2x \div -2 > 8 \div -2 \quad \leftarrow \text{flipped}$$

$$x > -4$$

Check: $x = 0 \rightarrow 0 < 8 \checkmark$

$x = -10 \rightarrow 20 < 8 \times$

Task 6.3 • One step

DO

#	INEQUALITY	SOLUTION	#	INEQUALITY	SOLUTION
a	$x + 6 > 10$		d	$x/3 \geq 2$	
b	$x - 3 \leq 5$		e	$-x > 3$	
c	$4x < 20$		f	$-5x \leq 15$	

Task 6.4 • Find the error

DETECTIVE

Ivy

$$-4x > 20$$

$$x > -5$$

What went wrong?

Correct answer: _____

Dev

$$x + 7 \leq 3$$

$$x \leq 10$$

What went wrong?

Correct answer: _____

Nell

$$x \geq 2 \text{ drawn with an open circle}$$

What went wrong?

Correct answer: _____

Two steps and real limits

STRETCH

19 MIN

Task 6.5 • Two steps

DO

Undo the adding and subtracting first, then the multiplying. Watch for the flip.

#	INEQUALITY	WORKING	SOLUTION	#	INEQUALITY	WORKING	SOLUTION
a	$2x + 5 > 11$			d	$x/2 - 3 \geq 1$		
b	$3x - 4 \leq 8$			e	$-3x + 1 > 7$		
c	$5x + 2 < -13$			f	$4 - 2x \leq 10$		

Two of these needed the flip. Which two, and what did they have in common? _____

Task 6.6 • Real limits

APPLY

Write the inequality first, then solve it, then answer in a sentence.

a. A ride costs \$6 and you have \$40. How many rides can you take?

c. A phone plan is \$25 a month plus 10c a text. You want to stay under \$40. How many texts?

b. A lift holds 500 kg. The operator weighs 80 kg and each box is 35 kg. How many boxes?

d. You need an average of at least 70 over four tests. You have 65, 72 and 68. What do you need on the fourth?

Task 6.7 • Think it through

CHALLENGE

a. Describe in words every number that fits $-3 \leq x \leq 5$. _____

b. Find one value of x that makes $x + 1 < x$ true. What does that tell you? _____

c. How many values of x make $x + 1 > x$ true? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

More Than, Less Than

ANSWERS

Task 6.1 • Into symbols

- a. $a \geq 13$
- b. $p \leq 8$
- c. $s < 20$
- d. $t > 5$
- e. $n > 60$
- f. $w \leq 23$

a and f use “at least” and “at most”. Both include the boundary number itself — 13 is old enough, and a 23 kg bag is allowed — so both get the line underneath.

Task 6.1b • Does it work?

$$x \leq 4 \rightarrow 4, 0, -7, 3.99$$

not 4.5, not 10

$$x > -2 \rightarrow 0, -1.9, 100$$

not -2 (not strictly greater),
not -5, not -2.1

Including 4 but not 4.5 is the whole idea of $\leq$. Students who leave out 4 have read it as $<$.

Task 6.2 • The four number lines

- a. $x > 1$ open circle at 1, arrow right
- b. $x \leq -3$ filled circle at -3, arrow left
- c. $x \geq 0$ filled circle at 0, arrow right
- d. $x < 5$ open circle at 5, arrow left

Mark it wrong only if the circle type or the arrow direction is wrong. Neatness is not the point.

Task 6.3 • One step

- a. $x > 4$
- b. $x \leq 8$
- c. $x < 5$
- d. $x \geq 6$
- e. $x < -3$ flipped
- f. $x \geq -3$ flipped

e and f both divided by a negative. In e, $-x > 3$ means x is less than -3 — test $x = -10$: $-(-10) = 10$, and $10 > 3$. ✓

Task 6.5 • Two steps

- a. $2x > 6 \rightarrow x > 3$
- b. $3x \leq 12 \rightarrow x \leq 4$
- c. $5x < -15 \rightarrow x < -3$
- d. $x/2 \geq 4 \rightarrow x \geq 8$
- e. $-3x > 6 \rightarrow x < -2$ flipped
- f. $-2x \leq 6 \rightarrow x \geq -3$ flipped

e and f are the two that flip. What they share is a negative number in front of the x at the point of dividing. Nothing else on the page flips.

Task 6.4 • The three errors

Ivy — $x < -5$. She divided by -4 and did not flip. Test $x = 0$: $-0 > 20$ is false, so 0 cannot be in the answer — which rules her version out immediately.

Dev — $x \leq -4$. He added 7 instead of subtracting it.

Nell — the circle must be filled. $\geq$ includes 2 itself.

Task 6.6 • Real limits

- a. $6r \leq 40 \rightarrow r \leq 6.67 \rightarrow 6$ rides
- b. $80 + 35b \leq 500 \rightarrow 35b \leq 420 \rightarrow 12$ boxes
- c. $25 + 0.1t < 40 \rightarrow t < 150 \rightarrow 149$ texts
- d. $65+72+68 = 205$, total needs 280 $\rightarrow$ at least 75

a and c are worth discussing. The algebra gives 6.67 and 150, but you cannot take two thirds of a ride, and 150 texts costs exactly \$40 rather than under it. The maths gives a boundary; the situation decides which side of it you land on. A student who writes 6.67 and stops has done the algebra and not the question.

Task 6.7 • Challenge

a. Every number from -3 up to 5, including both ends. On a line it is a filled circle at each end with the segment shaded between them — the first time they have seen an answer that is bounded on both sides.

b. There is no such value. A number plus one is always larger than the number. Some students will try several and conclude it is impossible, which is exactly the right conclusion. This is an inequality with *no* solution.

c. All of them. Every number satisfies it — infinitely many solutions. Put b and c side by side: same two expressions, opposite symbol, and between them they cover nothing and everything.

Boxes and Cans

2 MINUTES

Students calculate volume and surface area of rectangular boxes, discover that boxes holding exactly the same amount can need wildly different amounts of cardboard, unfold a net, then price up real packaging and fill a fish tank.

You do not need to know this subject

Two formulas, both printed on the student page. Volume is length $\times$ width $\times$ height. Surface area is the six faces added up. The key shows every calculation in full.

Photocopy these pages only

Pages 37–39

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Calculators are genuinely useful in this one — allow them if you have them.

WHAT TO SAY TO START

“A cereal box and a cube can hold exactly the same amount of cereal, and one of them costs the company nearly twice as much in cardboard. Today you work out which, and why every company on earth cares about the answer.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the two formulas	Volume fills, surface area wraps
12–26	Task 7.1 — the table of boxes	Steady practice, allow calculators
26–38	Task 7.2 — the net	Six faces, three pairs
38–50	Task 7.3 and 7.4 — packaging and the tank	Watch the units
50–60	Challenge, collect sheets	Discussion question below

Questions students will ask

“What is the difference between volume and surface area?”

Volume is how much fits inside — measured in cubic units. Surface area is how much wrapping paper you would need — measured in square units.

“Do I have to work out all six faces separately?”

No. Opposite faces are identical, so work out three and double each. That shortcut is the whole reason the formula has three doubled terms.

“Why cm^3 and not cm^2 ?”

Because you multiplied three lengths together. Two lengths give square units, three give cubic. Students who write cm^2 for a volume have usually done the arithmetic correctly — mark the number right and the unit wrong.

If they finish early

Task 7.6 asks for every whole-number box holding exactly 24 cm^3 , then which uses the least cardboard. There are six of them and the winner is the one closest to a cube.

Curriculum focus • for the regular teacher’s records

Grade 6–7: volume of right rectangular prisms, and surface area using nets. Discussion question for the last five minutes: “Why is almost every parcel you receive shaped roughly like a cube?”

Inside and outside

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Two questions about the same box

VOLUME – what fits inside

 $V = \text{length} \times \text{width} \times \text{height}$

$$6 \times 4 \times 3 = 72 \text{ cm}^3$$

three lengths → cubic units

SURFACE AREA – the wrapping

$$SA = 2(lw) + 2(lh) + 2(wh)$$

$$2(24) + 2(18) + 2(12) \\ = 48 + 36 + 24 = 108 \text{ cm}^2$$

two lengths → square units

Every box has **three pairs** of identical faces:

top and bottom
front and back
left end and right end

Work out three, double each one. Never six.

Task 7.1 • Fill in the table

DO

#	BOX (L × W × H)	VOLUME	SURFACE AREA
a	4 × 3 × 2		
b	6 × 5 × 2		
c	10 × 4 × 3		
d	5 × 5 × 5 (a cube)		
e	8 × 8 × 2		

All measurements are in centimetres. Write the units in every answer. Which box has the largest volume but not the largest surface area?

Task 7.1b • Same amount inside, different amount of cardboard

INVESTIGATE

All three of these boxes hold exactly 36 cm^3 . Work out how much cardboard each one needs.

BOX	VOLUME	SURFACE AREA	SHAPE OF IT
36 × 1 × 1	36 cm ³		
6 × 6 × 1	36 cm ³		
3 × 3 × 4	36 cm ³		

Which shape used the least cardboard, and what is it closest to? _____

Nets and packaging

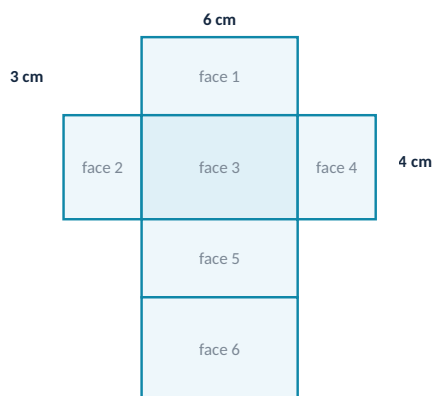
KEY SKILL

20 MIN

Task 7.2 • Unfold it

DO

This is the box from the panel, cut open and laid flat. It is 6 cm long, 4 cm wide and 3 cm high.



a. Write the dimensions of each face in the table.

FACE	SIZE	AREA
1		
2		
3		
4		
5		
6		

b. Total surface area? _____

c. Which faces are identical pairs?

d. Volume of the folded box? _____

Task 7.3 • What the cardboard costs

APPLY

Cardboard costs \$0.0004 per cm^2 . Work out what each package costs to make.

PACKAGE	SIZE (CM)	VOLUME	SURFACE AREA	COST
Cereal box	$30 \times 20 \times 8$			
Shoebox	$33 \times 20 \times 12$			
Tissue box	$24 \times 12 \times 10$			
Pizza box	$40 \times 40 \times 5$			

The shoebox holds more than the cereal box *and* costs more to make. Is the extra cost worth it per cm^3 of space?

Tanks, errors and the best box

STRETCH

22 MIN

Task 7.4 • The fish tank

APPLY

A glass tank is $90 \times 40 \times 50$ cm with no lid. $1,000 \text{ cm}^3 = 1$ litre, and a litre of water weighs 1 kg.

a. How many litres does it hold when full?

b. Filled to 5 cm below the top — how many litres?

c. What does the water in b weigh?

d. How much glass was needed? There is no lid.

Task 7.5 • Find the error

DETECTIVE

Reza

cube, side 4

$$V = 4 + 4 + 4 \\ = 12$$

What went wrong?

Correct answer: _____

Adabox $6 \times 4 \times 3$

$$SA = 6 \times 24 \\ = 144 \text{ cm}^2$$

What went wrong?

Correct answer: _____

Kofibox $4 \times 3 \times 2$

$$V = 24 \text{ cm}^2$$

What went wrong?

Correct answer: _____

Task 7.6 • Think it through

CHALLENGE

a. Find every box with whole-number sides that holds exactly 24 cm^3 . There are six. List them with the surface area of each.

b. Which one uses the least cardboard? _____

c. So why is a cereal box not a cube? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Boxes and Cans

ANSWERS

Task 7.1 • The table

- a. $4 \times 3 \times 2$ $V = 24 \text{ cm}^3$ $SA = 2(12) + 2(8) + 2(6) = 52 \text{ cm}^2$
 b. $6 \times 5 \times 2$ $V = 60 \text{ cm}^3$ $SA = 2(30) + 2(12) + 2(10) = 104 \text{ cm}^2$
 c. $10 \times 4 \times 3$ $V = 120 \text{ cm}^3$ $SA = 2(40) + 2(30) + 2(12) = 164 \text{ cm}^2$
 d. $5 \times 5 \times 5$ $V = 125 \text{ cm}^3$ $SA = 6 \times 25 = 150 \text{ cm}^2$
 e. $8 \times 8 \times 2$ $V = 128 \text{ cm}^3$ $SA = 2(64) + 2(16) + 2(16) = 192 \text{ cm}^2$

Draw out **d** against **e**: the $8 \times 8 \times 2$ box holds 3 cm^3 more than the cube and needs 42 cm^2 more cardboard to do it.

Task 7.1b • Same 36 cm^3

- $36 \times 1 \times 1$ $SA = 146$ a stick
 $6 \times 6 \times 1$ $SA = 96$ a slab
 $3 \times 3 \times 4$ $SA = 66$ near-cube

$3 \times 3 \times 4$ wins — the one closest to a cube. All three hold the same, yet the stick needs more than twice the cardboard. This is the central idea of the lesson.

Task 7.5 • The three errors

Reza — 64 cm^3 . Added, not multiplied.

Ada — 108 cm^2 . One face $\times 6$ only works for a cube.

Kofi — 24 cm^3 . Number right, unit wrong.

Task 7.2 • The net

- 1 $6 \times 3 = 18$ 3 $6 \times 4 = 24$ 5 $6 \times 3 = 18$
 2 $4 \times 3 = 12$ 4 $4 \times 3 = 12$ 6 $6 \times 4 = 24$
 b. total = 108 cm^2 d. $6 \times 4 \times 3 = 72 \text{ cm}^3$

c. Pairs are **1 and 5**, **2 and 4**, **3 and 6**.

Task 7.3 • Packaging costs

Cereal	$30 \times 20 \times 8$	$V \ 4,800 \text{ cm}^3$	$SA \ 2,000 \text{ cm}^2$	$\$0.80$
Shoebbox	$33 \times 20 \times 12$	$V \ 7,920 \text{ cm}^3$	$SA \ 2,592 \text{ cm}^2$	$\$1.04$
Tissue	$24 \times 12 \times 10$	$V \ 2,880 \text{ cm}^3$	$SA \ 1,296 \text{ cm}^2$	$\$0.52$
Pizza	$40 \times 40 \times 5$	$V \ 8,000 \text{ cm}^3$	$SA \ 4,000 \text{ cm}^2$	$\$1.60$

Yes — the shoebbox is better value: about $\$0.000131$ per cm^3 against $\$0.000167$, because it is the chunkier shape. The pizza box is the cautionary tale — it holds more than the shoebbox yet costs half as much again, because a flat shape is the worst way to enclose a volume.

Task 7.4 • The fish tank

- a. $90 \times 40 \times 50 = 180 \text{ L}$
 b. $90 \times 40 \times 45 = 162 \text{ L}$
 c. **162 kg**
 d. base $3,600$
 sides $9,000$
 ends $4,000$
 = **$16,600 \text{ cm}^2$**

In b, watch for 5 taken off the wrong dimension. A lid included in d gives 20,200 — partial credit.

Task 7.6 • Challenge

- $24 \times 1 \times 1$ **98** $6 \times 4 \times 1$ **68**
 $12 \times 2 \times 1$ **76** $6 \times 2 \times 2$ **56**
 $8 \times 3 \times 1$ **70** $4 \times 3 \times 2$ **52** ←

b. $4 \times 3 \times 2$, at 52 cm^2 — again the one closest to a cube. Accept five of six, and rotations as the same box.

c. Cardboard is not the only cost. A cube cereal box will not fit a shelf, is harder to pour from, and shows a smaller front panel. Any real constraint is correct.

Angles Everywhere

READ BEFORE THE BELL

2 MINUTES

Students name angle types, find missing angles on straight lines and around points, use the parallel line rules, then apply the lot to a roof truss and a road junction. The easiest lesson in the pack to supervise — almost every answer is a subtraction.

You do not need to know this subject

Four facts cover the whole lesson and all four are printed on the student page: a straight line makes 180° , a full turn makes 360° , a triangle makes 180° , and vertically opposite angles are equal.

Photocopy these pages only

Pages 42–44

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A protractor is not needed — every angle is calculated, not measured.

WHAT TO SAY TO START

"There is no protractor on this sheet, and you will not need one. Every angle on these pages can be worked out from the ones you are given. That is what an engineer does when the drawing is incomplete."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the four facts panel	They are all you need
12–24	Task 8.1 — name them and find them	Fast, confidence-building
24–36	Task 8.2 — parallel lines	Eight angles, only two values
36–48	Tasks 8.3 and 8.4 — truss and junction	Real drawings
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Do we need protractors?"

No. Nothing on the sheet is measured. If a student starts measuring, they will get slightly wrong answers and think the sheet is broken.

"How do I check a parallel-lines answer without knowing the rules?"

There are only two values on the whole diagram, and they add to 180. Any answer that is not 65 or 115 is wrong.

"What is a reflex angle?"

Anything bigger than 180° and less than 360° — the long way round. Task 8.5 has a student who calls 130° reflex, which it is not.

If they finish early

Task 8.6 asks whether a triangle can have two right angles. It cannot, and working out why is a much better use of five minutes than more subtraction.

Curriculum focus • for the regular teacher's records

Grade 7–8: complementary, supplementary, vertical and adjacent angles; the angle sum of a triangle; angles formed by parallel lines cut by a transversal. Discussion question for the last five minutes: *"Why can a triangle never have two right angles?"*

Naming and finding

UNDERSTAND

16 MIN

NAME _____

CLASS _____

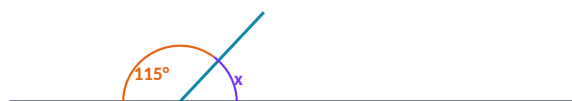
DATE _____

Four facts, and everything else follows

STRAIGHT LINE = 180°
FULL TURN = 360°
TRIANGLE = 180°
OPPOSITE ANGLES EQUAL

acute under 90° straight 180°
 right exactly 90° reflex over 180°
 obtuse 90° to 180°

On a straight line. The two angles must add to 180° , so $x = 180 - 115 = 65^\circ$.



Complementary $\rightarrow 90^\circ$. **Supplementary** $\rightarrow 180^\circ$.

Task 8.1 • Name the angle

DO

Write acute, right, obtuse, straight or reflex beside each one.

- a. 35° _____ b. 90° _____ c. 128° _____
 d. 180° _____ e. 295° _____ f. 89° _____

Task 8.1b • Find the missing angle

DO

#	THE SITUATION	x
a	Two angles on a straight line: 115° and x	
b	Angles round a point: 120° , 155° and x	
c	A triangle with angles 60° , 50° and x	
d	A right-angled triangle with angles 90° , 35° and x	
e	x is vertically opposite an angle of 100°	
f	Three angles on a straight line: 40° , 75° and x	

Task 8.1c • Complement and supplement

DO

ANGLE	30°	45°	72°	15°
Complement				
Supplement				

Parallel lines and triangles

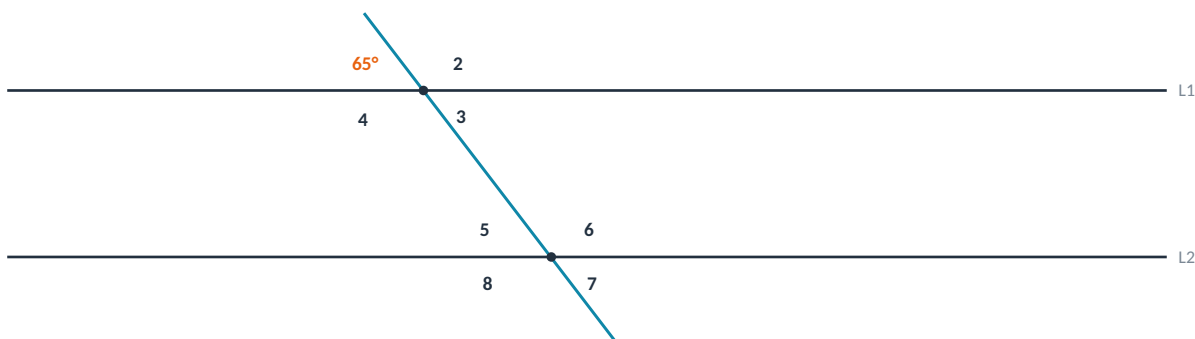
KEY SKILL

20 MIN

Task 8.2 • Parallel lines

DO

L1 and L2 are parallel. One angle is given. Fill in all eight.



ANGLE	SIZE	ANGLE	SIZE
1	65° • given	5	
2		6	
3		7	
4		8	

Only two different numbers appear in all eight boxes. What do they add up to? _____

Task 8.2b • Triangles

DO

#	THE TRIANGLE	MISSING ANGLES
a	Isosceles, apex angle 40° . Find the two base angles.	
b	Right-angled, one other angle 28° . Find the third.	
c	Two angles are 55° and 65° . Find the third, and the exterior angle beside it.	

Task 8.3 • Find the error

DETECTIVE

Sofia

triangle angles
 80° , 60° , 50°

What went wrong?

Correct answer: _____

Marc

"They are on a
straight line so
they must be equal"

What went wrong?

Correct answer: _____

Nia

" 130° is a
reflex angle"

What went wrong?

Correct answer: _____

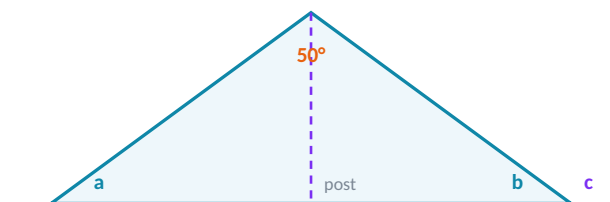
Trusses, junctions and proof

STRETCH

19 MIN

Task 8.4 • The roof truss

APPLY

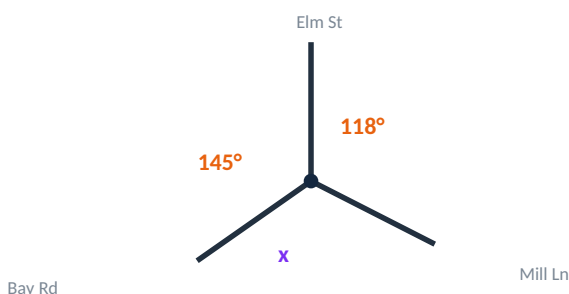


The truss is an isosceles triangle. The apex angle is 50° and the two sloping sides are equal. The dashed post is vertical.

- Angle a? _____
- Angle b? _____
- Angle c, outside the truss? _____
- The post cuts the truss in half. Write the three angles of the left-hand half. _____

Task 8.5 • The road junction

APPLY



Three roads meet at a single point. Two of the angles between them are marked.

- Find x. _____
- Which fact did you use, 180° or 360° ? _____
- A fourth road is added inside the 145° angle, splitting it in the ratio 2:3. What are the two new angles? _____

Task 8.6 • Think it through

CHALLENGE

- Why does an obtuse angle have a supplement but never a complement? _____
- A triangle has angles x, x and $3x$. Find all three. _____
- Why can a triangle never have two right angles? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

 Teacher initials _____

Answer key

Angles Everywhere

ANSWERS

Task 8.1 • Name the angle

- a. 35° **acute** d. 180° **straight**
 b. 90° **right** e. 295° **reflex**
 c. 128° **obtuse** f. 89° **acute**

f catches people who round: 89° is under 90° , so it is acute.

Task 8.1c • Complement and supplement

	30°	45°	72°	15°
comp	60	45	18	75
supp	150	135	108	165

45° is its own complement. An obtuse angle is already over 90° , so there is nothing left to add — the complement would have to be negative.

Task 8.1b • Missing angles

- a. $180 - 115 = 65^\circ$ d. $180 - 90 - 35 = 55^\circ$
 b. $360 - 120 - 155 = 85^\circ$ e. opposite angles = **100°**
 c. $180 - 60 - 50 = 70^\circ$ f. $180 - 40 - 75 = 65^\circ$

Only e needs a rule rather than a subtraction, and the rule is that vertically opposite angles are simply equal — no working required.

Task 8.2 • Parallel lines

- 1 **65°** 5 **65°**
 2 **115°** 6 **115°**
 3 **65°** 7 **65°**
 4 **115°** 8 **115°**

Only two values appear and they add to **180**. Anything that is not 65 or 115 is wrong, so students can check their own eight boxes in five seconds.

Task 8.2b • Triangles

- a. $180 - 40 = 140$, halved → **70° and 70°**
 b. $180 - 90 - 28 = 62^\circ$
 c. $180 - 55 - 65 = 60^\circ$
 exterior $180 - 60 = 120^\circ$

In c the exterior angle equals the two far interior angles added: $55 + 65 = 120$. Students who spot that have found a real theorem.

Task 8.3 • The three errors

Sofia — they add to 190, not 180. One angle is 10° too big; the triangle as described cannot exist.

Marc — no. Angles on a straight line add to 180° . They are only equal in the special case where both are 90° .

Nia — 130° is obtuse. Reflex means over 180° .

Task 8.4 • The roof truss

- a. $(180 - 50) \div 2 = 65^\circ$
 b. equal to a = **65°**
 c. $180 - 65 = 115^\circ$
 d. **$90^\circ, 65^\circ, 25^\circ$**

In d the post is vertical so it makes 90° with the base; the base angle is still 65° ; the apex has been halved to 25° . They add to 180, which is the check.

Task 8.5 • The road junction

- a. $360 - 145 - 118 = 97^\circ$
 b. the **360°** fact — angles round a point
 c. $145 \div 5 = 29$
 $2 \times 29 = 58^\circ$
 $3 \times 29 = 87^\circ$

In c the two parts must add back to 145, which is the check. A student who splits into 2 and 3 degrees has read ratio as amount.

Task 8.6 • Challenge

a. An obtuse angle is already over 90° , so its complement would have to be a negative angle — there is nothing left to add. Its supplement is fine, because 180° leaves room to spare.

b. $x + x + 3x = 5x = 180$, so $x = 36$. The angles are **$36^\circ, 36^\circ$ and 108°** .

c. Two right angles already make 180° , leaving 0° for the third. A 0° angle means the two sides lie on top of each other, so there is no triangle. Accept any answer that gets to “nothing left for the third angle”.

The Diagonal Problem

READ BEFORE THE BELL

2 MINUTES

Students use $a^2 + b^2 = c^2$ to find hypotenuses and missing legs, test whether a triangle is right-angled, then apply it to a ladder, a television, a shortcut across a park and a gate brace. Finishes with distances on a grid.

You do not need to know this subject

One formula, printed on the student page. The only judgement call is whether you are adding (to find the long side) or subtracting (to find a short one), and the key says which for every question.

Photocopy these pages only

Pages 47–49

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator with a square-root key makes this much smoother — allow one if you have it.

WHAT TO SAY TO START

"You cannot measure the diagonal of a room by walking through the wall. But if you know two sides of a right angle, you can always calculate the third — and people have been doing it for four thousand years."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the formula panel	c is always the longest side
12–24	Task 9.1 — hypotenuse and legs	Add for c, subtract for a leg
24–34	Task 9.2 — is it right-angled?	Test, do not assume
34–48	Task 9.3 — real problems	Draw the triangle first
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Which side is c?"

Always the one opposite the right angle, and always the longest. If a student has c as a short side, everything downstream will be wrong.

"Do I add or subtract?"

Add when the missing side is the long one. Subtract when the missing side is a short one. A quick check: your answer for a short side must come out smaller than c.

"What if the square root is not a whole number?"

That is fine and it happens twice on the sheet. Round to two decimal places, or leave it as a square root if the student prefers.

If they finish early

Task 9.6 asks whether a 210 cm sheet of plywood will fit through a doorway 80 cm by 200 cm. It will, but only diagonally, and only just.

Curriculum focus • for the regular teacher's records

Grade 8: applying the Pythagorean theorem to find unknown side lengths, to determine whether a triangle is right-angled, and to find distances between points on a coordinate grid. Discussion question for the last five minutes: *"Why does the formula only work for right-angled triangles?"*

One formula, two directions

UNDERSTAND

16 MIN

NAME _____

CLASS _____

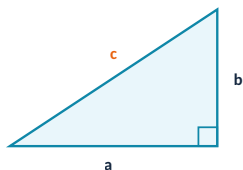
DATE _____

The oldest useful formula in mathematics

$$a^2 + b^2 = c^2$$

c is the **hypotenuse** – opposite the right angle, and always the longest side.

a and **b** are the two short sides, the legs.



FINDING c – add

$$3^2 + 4^2 = 9 + 16 = 25$$

$$c = \sqrt{25} = 5$$

FINDING A LEG – subtract

$$c = 13, a = 5$$

$$b^2 = 169 - 25 = 144$$

$$b = \sqrt{144} = 12$$

Task 9.1 • Find the hypotenuse

DO

Round to two decimal places if it is not a whole number.

#	A	B	c
a	3	4	
b	9	12	
c	5	12	

#	A	B	c
d	8	15	
e	7	24	
f	2	3	

Task 9.1b • Find the missing leg

DO

Here you are given the hypotenuse and one leg.

#	C	A	B
a	10	6	
b	25	7	
c	17	8	

#	C	A	B
d	13	12	
e	13	5	
f	15	9	

Testing and applying

KEY SKILL

20 MIN

Task 9.2 • Is it right-angled?

TEST

Square the two short sides, add them, and see whether you get the square of the longest. Write yes or no, and show the numbers.

#	SIDES	$A^2 + B^2$	C^2	RIGHT-ANGLED?
a	6, 8, 10			
b	5, 6, 8			
c	9, 40, 41			
d	4, 5, 6			
e	10, 24, 26			
f	7, 8, 11			

Three of these are right-angled. What would go wrong if a builder assumed all six were? _____

Task 9.3 • Real problems

APPLY

Sketch the triangle first. Label the right angle before you do any arithmetic.

a. A 10 ft ladder rests against a wall with its foot 6 ft out. How high up the wall does it reach?

c. You walk 300 m east then 400 m north. How far are you from the start, and how much would the direct path have saved?

e. A wheelchair ramp rises 1.5 m over a horizontal run of 3.6 m. How long is the ramp surface itself?

b. A television screen is 48 in wide and 27 in high. Screens are sold by their diagonal — what size is it?

d. A gate is 3 ft wide and 4 ft tall. How long is the diagonal brace?

f. A field is 120 m by 160 m. How much shorter is the diagonal path than walking around two sides?

Grids, errors and a tight squeeze

STRETCH

19 MIN

Task 9.4 • Distance on a grid

APPLY

The difference in x is one leg. The difference in y is the other. The distance between the points is the hypotenuse.

#	FROM	TO	ACROSS	UP	DISTANCE
a	(1, 2)	(4, 6)			
b	(0, 0)	(5, 12)			
c	(-2, 1)	(6, 16)			
d	(3, 7)	(3, 17)			

One of these did not need the formula at all. Which one, and why? _____

Task 9.5 • Find the error

DETECTIVE

Bea

legs 5 and 12
 $c = 5 + 12 = 17$

What went wrong?

Correct answer: _____

Cal

$c = 13$, $a = 12$
 $b^2 = 169 + 144$
 $b = 17.69$

What went wrong?

Correct answer: _____

Dee

"3, 4, 5 works, so
 6, 7, 8 must too"

What went wrong?

Correct answer: _____

Task 9.6 • Think it through

CHALLENGE

- a. A square has sides of 6 cm. How long is its diagonal? _____
- b. A doorway is 80 cm wide and 200 cm tall. Will a rigid sheet of plywood 210 cm long fit through it? Show the working.
- _____
- _____
- c. Find a set of three whole numbers, other than 3-4-5, that makes a right-angled triangle. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

The Diagonal Problem

ANSWERS

Task 9.1 • Hypotenuse

- a. 3, 4 $9 + 16 = 25$ $c = 5$
 b. 9, 12 $81 + 144 = 225$ $c = 15$
 c. 5, 12 $25 + 144 = 169$ $c = 13$
 d. 8, 15 $64 + 225 = 289$ $c = 17$
 e. 7, 24 $49 + 576 = 625$ $c = 25$
 f. 2, 3 $4 + 9 = 13$ $c = 3.61$

f is the one that does not come out whole. $\sqrt{13} = 3.6055... \rightarrow 3.61$. Accept $\sqrt{13}$ left as it is.

Task 9.1b • Missing leg

- a. $100 - 36 = 64$ $b = 8$
 b. $625 - 49 = 576$ $b = 24$
 c. $289 - 64 = 225$ $b = 15$
 d. $169 - 144 = 25$ $b = 5$
 e. $169 - 25 = 144$ $b = 12$
 f. $225 - 81 = 144$ $b = 12$

Every answer here is smaller than the given c, which is the fastest way for a student to catch an error — a leg longer than the hypotenuse is impossible.

Task 9.2 • Is it right-angled?

- a. 6, 8, 10 $36 + 64 = 100$ 100 **yes**
 b. 5, 6, 8 $25 + 36 = 61$ 64 **no**
 c. 9, 40, 41 $81 + 1600 = 1681$ 1681 **yes**
 d. 4, 5, 6 $16 + 25 = 41$ 36 **no**
 e. 10, 24, 26 $100 + 576 = 676$ 676 **yes**
 f. 7, 8, 11 $49 + 64 = 113$ 121 **no**

b and f look right in a sketch, which is the point — a builder who assumed all six were square would get walls that do not meet and doors that will not close. Note that a is 3-4-5 doubled and e is 5-12-13 doubled: scaling keeps a right triangle right.

Task 9.3 • Real problems

- a. $100 - 36 = 64 \rightarrow 8$ ft up the wall
 b. $2304 + 729 = 3033$
 $\sqrt{3033} = 55.07 \rightarrow$ a **55-inch** screen
 c. $90000 + 160000 = 250000$
 $\rightarrow$ **500 m**; saves **200 m**
 d. $9 + 16 = 25 \rightarrow$ brace is **5 ft**
 e. $2.25 + 12.96 = 15.21$
 $\rightarrow$ ramp is **3.9 m**
 f. $14400 + 25600 = 40000$
 diagonal **200 m** vs 280 m
 $\rightarrow$ saves **80 m**

b is worth a comment: this is exactly how televisions are advertised, and it is why a 55-inch screen is nothing like 55 inches wide.

Task 9.4 • Distance on a grid

- a. across 3, up 4 $\rightarrow 5$
 b. across 5, up 12 $\rightarrow 13$
 c. across 8, up 15 $\rightarrow 17$
 d. across 0, up 10 $\rightarrow 10$

d needed no formula — both points have $x = 3$, so just subtract: $17 - 7 = 10$. That shortcut only works when the points share an x or a y . In c, watch for across written as 4 instead of 8; -2 to 6 is eight steps.

Task 9.5 • The three errors

Bea — 13. She added the sides instead of the squares. $25 + 144 = 169$, and $\sqrt{169} = 13$.

Cal — 5. He is looking for a leg, so he should subtract: $169 - 144 = 25$. His answer of 17.69 is longer than the hypotenuse, which is impossible.

Dee — no. $36 + 49 = 85$, not 64. Scaling works only when you multiply every side by the same number: 6-8-10 works, 6-7-8 does not.

Task 9.6 • Challenge

a. $36 + 36 = 72$, so the diagonal is $\sqrt{72} \approx 8.49$ cm.

b. **Yes, but only diagonally and only just.** $80^2 + 200^2 = 6,400 + 40,000 = 46,400$, and $\sqrt{46,400} \approx 215.4$ cm. The sheet is 210 cm, so it clears by about 5 cm — straight through the doorway it would not fit at all, since 210 is more than 200.

c. 5-12-13, 8-15-17, 7-24-25, 9-40-41 and 6-8-10 all work. Accept any triple the student verifies.

Rate of Change

2 MINUTES

Students find the rate of change in a table, calculate slope between two points, read two lines off a graph, turn word situations into equations, and finish by comparing two phone plans to find the exact minute at which the cheaper one changes.

You do not need to know this subject

Every rule has the same shape: $y = (\text{rate}) \times x + (\text{starting value})$. The key gives the rule and the answer for every question, and every table can be checked by substituting a number back in.

Photocopy these pages only

Pages 52–54

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler. A calculator is helpful for Task 10.4.

WHAT TO SAY TO START

“Two phone plans. One is a flat \$40 a month, the other is \$15 plus five cents a minute. Which is cheaper depends entirely on how much you talk — and today you find the exact minute where the answer flips.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the slope panel	Rise over run, and what b means
14–26	Task 10.1 — tables and slopes	Table D is the trap
26–38	Task 10.2 — reading the graph	Two lines, one crossing point
38–50	Task 10.3 — stories into equations	Rate first, then starting value
50–60	Task 10.4 and challenge, collect sheets	Discussion question below

Questions students will ask

“What is slope, in plain words?”

How much y changes every time x goes up by one. On a graph it is the steepness; in a story it is the rate — dollars per month, centimetres per year.

“What does the b in $y = mx + b$ mean?”

Where you start, before any x has happened. On a graph it is where the line crosses the vertical axis. In a story it is the joining fee, the starting height, the full tank.

“Can a slope be negative or zero?”

Yes to both. Negative means it goes down as x increases — a draining tank. Zero means it is flat and never changes, like a fixed monthly fee.

If they finish early

Task 10.6 asks for two lines that never meet. Any two with the same slope and different starting values will do, and students usually find that surprising.

Curriculum focus • for the regular teacher’s records

Grade 8: understanding slope as a constant rate of change, deriving $y = mx + b$, and comparing proportional and linear relationships represented in tables, graphs and equations. Discussion question for the last five minutes: “Where in the room can you see something changing at a constant rate?”

Rates in tables and points

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Every straight line has the same shape

$$y = mx + b$$

m is the **slope** – how much **y** changes each time **x** goes up by one

b is where you **start**, the value when **x** = 0

SLOPE FROM TWO POINTS

$$m = \frac{\text{change in } y}{\text{change in } x}$$

$$\begin{aligned} &(1, 4) \text{ and } (3, 10) \\ m &= (10 - 4) \div (3 - 1) \\ &= 6 \div 2 = 3 \end{aligned}$$

READING A TABLE

x	0	1	2	3
y	5	8	11	14

y goes up **3** each time, and starts at **5**

$$y = 3x + 5$$

Task 10.1 • Find the rule

DO

Work out the rule for each table, then use it to predict **y** when **x** = 10.

#	X VALUES	Y VALUES	RULE	Y AT X = 10
A	0, 1, 2, 3	2, 6, 10, 14		
B	0, 1, 2, 3	10, 8, 6, 4		
C	0, 1, 2, 3	0, 7, 14, 21		
D	0, 2, 4, 6	3, 9, 15, 21		

One of these tables does not step up by 1 in the **x** column. Which, and why does that change the working?

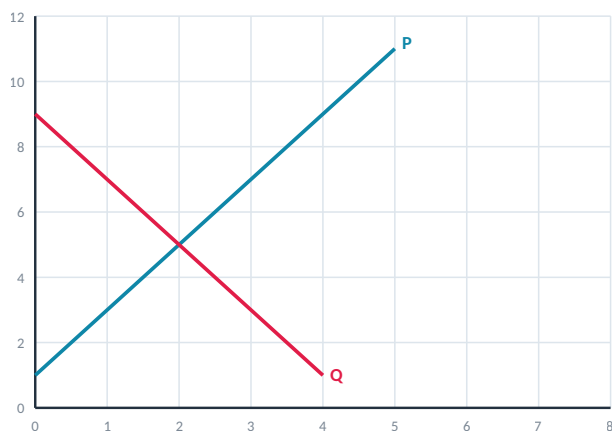
Task 10.1b • Slope between two points

DO

#	TWO POINTS	SLOPE	#	TWO POINTS	SLOPE
a	(0, 1) and (2, 7)		d	(2, 5) and (6, 5)	
b	(1, 3) and (4, 9)		e	(1, 2) and (5, 12)	
c	(0, 8) and (3, 2)				

Task 10.2 • Read the graph

DO



Two lines are drawn. Work out the rule for each and the point where they meet.

- Where does P cross the vertical axis? _____
- What is P's slope? _____
- Write P's rule. _____
- Write Q's rule. _____
- Where do the two lines cross? _____
- Which line is falling, and how can you tell from its rule alone?

Task 10.3 • Stories into equations

APPLY

Find the rate first, then the starting value, then write the rule.

#	THE SITUATION	EQUATION	THEN ANSWER THIS
a	A tree is 120 cm tall and grows 15 cm a year.		How tall after 6 years?
b	A tank holds 400 L and drains at 8 L an hour.		When is it empty?
c	A gym charges \$30 to join, then \$12 a month.		Cost of a year?
d	A taxi charges \$5, then \$2.40 per km.		Cost of a 12 km trip?

Task 10.3b • Read the rule at a glance

THINK

You should be able to answer all four of these without doing any arithmetic.

$$y = 2x + 5 \quad y = 2x \quad y = 5x + 2 \quad y = -2x + 5$$

- Which two are parallel? _____
- Which one passes through the origin? _____
- Which one falls as x increases? _____
- Which one is steepest? _____

Phone plans and parallel lines

STRETCH

22 MIN

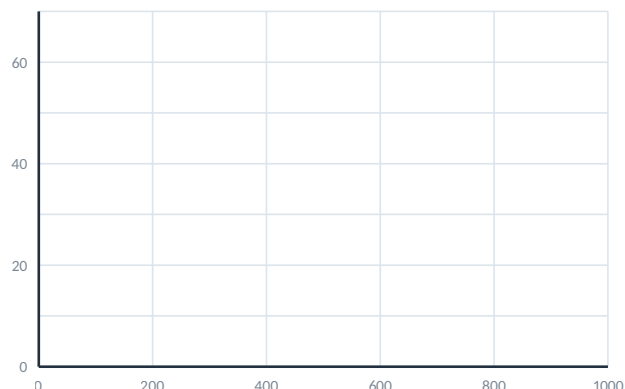
Task 10.4 • Two phone plans

APPLY

Plan A is a flat \$40 a month, however much you talk. **Plan B** is \$15 a month plus 5 cents a minute.

MINUTES	PLAN A	PLAN B
0		
200		
400		
600		
800		

Plot both plans. Minutes across the bottom, dollars up the side.



- Write a rule for each plan. _____
- At exactly how many minutes do they cost the same?

- Which plan should a light user choose? _____

Task 10.5 • Find the error

DETECTIVE

Ollie

$$y = 4x + 9$$

“the slope is 9”

What went wrong?

Correct answer: _____

Pia

(1,2) to (5,12)

$$m = 4 \div 10 = 0.4$$

What went wrong?

Correct answer: _____

Quinn

“the line goes down
so it has no slope”

What went wrong?

Correct answer: _____

Task 10.6 • Think it through

CHALLENGE

- Write two rules for lines that will never meet, and explain why. _____
- A line passes through (0, 6) and (3, 0). Write its rule. _____
- A line has slope 5 and passes through (2, 13). Find b. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Rate of Change

ANSWERS

Task 10.1 • Find the rule

- A $y = 4x + 2$ at $x=10 \rightarrow 42$
 B $y = -2x + 10$ at $x=10 \rightarrow -10$
 C $y = 7x$ at $x=10 \rightarrow 70$
 D $y = 3x + 3$ at $x=10 \rightarrow 33$

D is the trap. Its x column steps up by 2, not 1, so the y jump of 6 is two steps of the slope. The slope is 3, not 6. C has no starting value because it begins at zero — that is a proportional relationship.

Task 10.1b • Slope between points

- a. $6 \div 2 = 3$
 b. $6 \div 3 = 2$
 c. $-6 \div 3 = -2$
 d. $0 \div 4 = 0$
 e. $10 \div 4 = 2.5$

d is flat: y never changes, so the slope is zero. A slope of zero is a real answer, not a missing one — it is what a fixed monthly fee looks like.

Task 10.2 • Read the graph

- a. P crosses at 1
 b. slope of P = 2
 c. P: $y = 2x + 1$
 d. Q: $y = -2x + 9$
 e. they cross at (2, 5)

f. **Q is falling, and the minus sign in front of the 2 is the giveaway.** Check e by substituting: $2(2) + 1 = 5$ and $-2(2) + 9 = 5$. Both give 5, which is what a crossing point means.

Task 10.3 • Stories into equations

- a. $h = 15y + 120$ 6 yrs $\rightarrow$ 210 cm
 b. $V = -8t + 400$ empty at 50 h
 c. $C = 12m + 30$ 1 yr $\rightarrow$ \$174
 d. $C = 2.4k + 5$ 12 km $\rightarrow$ \$33.80

b is the only negative rate, because the tank is losing water. Setting $V = 0$ gives $8t = 400$, so $t = 50$.

Task 10.3b: a. $y = 2x + 5$ and $y = 2x$ — same slope. b. $y = 2x$, the only one with no starting value. c. $y = -2x + 5$. d. $y = 5x + 2$, the largest slope.

Task 10.4 • Two phone plans

minutes	0	200	400	600	800
Plan A	\$40	\$40	\$40	\$40	\$40
Plan B	\$15	\$25	\$35	\$45	\$55

- a. A: $C = 40$ B: $C = 0.05m + 15$
 b. $0.05m + 15 = 40$
 $0.05m = 25 \rightarrow m = 500$ minutes
 c. a light user should take **Plan B**

Plan A is the flat line; Plan B is the sloping one, and they cross at 500 minutes and \$40. Below 500 minutes B is cheaper, above it A is. Plan A having a slope of zero is the whole reason it can start out expensive and end up the bargain.

Task 10.5 • The three errors

Ollie — the slope is 4. The 9 is where the line starts, the value when $x = 0$.

Pia — 2.5. She divided the change in x by the change in y . Slope is change in y over change in x : $10 \div 4$.

Quinn — it has a negative slope. Falling is a slope, not the absence of one. Only a flat line has slope zero.

Task 10.6 • Challenge

- a. Any two rules with the **same slope and different starting values** — $y = 3x + 1$ and $y = 3x + 8$, for instance. They climb at exactly the same rate, so the gap between them never closes. These are parallel lines.
 b. $y = -2x + 6$. It starts at 6, and drops 6 over a run of 3, so the slope is -2 .
 c. $b = 3$, so $y = 5x + 3$. Substitute: $13 = 5(2) + b$ gives $13 = 10 + b$.

Thank you for covering this class. Please spend two minutes on this page before you leave and put it on the desk — it is the only handover the regular teacher will get.

SUBSTITUTE'S NAME

DATE

Class / period

Room

WHICH LESSON DID YOU USE?

- | | |
|--|--|
| ☐ Lesson 01 • Factors, GCF & LCM | ☐ Lesson 02 • Fraction operations |
| ☐ Lesson 03 • Proportion & scale drawings | ☐ Lesson 04 • Exponents & scientific notation |
| ☐ Lesson 05 • Expressions & like terms | ☐ Lesson 06 • Inequalities |
| ☐ Lesson 07 • Volume & surface area | ☐ Lesson 08 • Angle relationships |
| ☐ Lesson 09 • The Pythagorean theorem | ☐ Lesson 10 • Slope & linear relationships |

How far did the class get?

- ☐ Finished everything, including the challenge
- ☐ Finished the first two pages
- ☐ Finished the first page only
- ☐ Did not get started — see notes below

The sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Marked together in class using the key

ANYTHING THE TEACHER SHOULD KNOW

STUDENTS WHO WORKED ESPECIALLY WELL

STUDENTS WHO NEEDED MORE SUPPORT

From one teacher to another

Covering a class you do not teach, for students you do not know, from a folder you have never seen is genuinely hard. If today was messy, that is not a reflection on you. Write down what happened, leave the page here, and go home. Thank you.

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Standards alignment

Written against the Grade 6–8 band of the Common Core State Standards for Mathematics, and readily mapped to state equivalents and to the Australian Curriculum Years 7–8.

- **01 • The Number Detective** — Factors, GCF & LCM
- **02 • Slices and Shares** — Fraction operations
- **03 • Scale It Up** — Proportion & scale drawings
- **04 • Powers of Ten** — Exponents & scientific notation
- **05 • Same Thing, Shorter** — Expressions & like terms
- **06 • More Than, Less Than** — Inequalities
- **07 • Boxes and Cans** — Volume & surface area
- **08 • Angles Everywhere** — Angle relationships
- **09 • The Diagonal Problem** — The Pythagorean theorem
- **10 • Rate of Change** — Slope & linear relationships

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